

Is There a Competitive Advantage to Using Multivariate Statistical or Machine Learning Methods Over the Bross Formula in the hdPS Framework for Bias and Variance Estimation?

--Manuscript Draft--

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Full Title:	Is There a Competitive Advantage to Using Multivariate Statistical or Machine Learning Methods Over the Bross Formula in the hdPS Framework for Bias and Variance Estimation?
Short Title:	High-Dimensional Propensity Score
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Keywords:	Machine Learning; Propensity score; Causal inference; Confounding; Plasmode simulation; Inverse-probability weighting; epidemiological methods; Health data science
Abstract:	**Purpose**: We aim to evaluate various proxy selection methods within the context of high-dimensional propensity score (hdPS) analysis. This study aimed to systematically evaluate and compare the performance of traditional statistical methods and machine learning approaches within the hdPS framework, focusing on key metrics such as bias, standard error (SE), and coverage, under various exposure and outcome prevalence scenarios. **Methods**: We conducted a plasmode simulation study using data from the National Health and Nutrition Examination Survey (NHANES) cycles from 2013 to 2018. We compared methods including the kitchen sink model, Bross-based hdPS, Hybrid hdPS, LASSO, Elastic Net, Random Forest, XGBoost, and Genetic Algorithm (GA). The performance of each inverse probability weighted method was assessed based on bias, MSE, coverage probability, and SE estimation across three epidemiological scenarios: frequent exposure and outcome, rare exposure and frequent outcome, and frequent exposure and rare outcome. **Results**: XGBoost consistently demonstrated strong performance in terms of MSE and coverage, making it effective for scenarios prioritizing precision. However, it exhibited higher bias, particularly in rare exposure scenarios, suggesting it is less suited when minimizing bias is critical. In contrast, GA showed significant limitations, with consistently high bias and MSE, making it the least reliable method. Bross-based hdPS, and Hybrid hdPS methods provided a balanced approach, with low bias and moderate MSE, though coverage varied depending on the scenario. Rare outcome scenarios generally resulted in lower MSE and better precision, while rare exposure scenarios were associated with higher bias and MSE. Notably, traditional statistical approaches such as forward selection and backward elimination performed comparably to more sophisticated machine learning methods in terms of bias and coverage, suggesting that these simpler approaches may be viable alternatives due to their computational efficiency. **Conclusion**: The results highlight the importance of selecting hdPS methods based on the specific characteristics of the data, such as exposure and outcome prevalence. While advanced machine learning methods such as XGBoost can enhance precision, simpler methods such as forward selection or backward elimination may offer similar performance in terms of bias and coverage with fewer computational demands. Tailoring the choice of method to the epidemiological scenario is essential for optimizing the balance between bias reduction and precision.
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NHANES data is publicly accessible and can be retrieved from the NHANES website. The datasets generated and/or analyzed during the current study are available in the NHANES repository: <https://www.cdc.gov/nchs/nhanes/index.htm>. Analysis data and codes are provided in <https://github.com/ehsanx/hdPS-proxy-select-codes>, with a stable, citable release archived at Zenodo (DOI: 10.5281/zenodo.15201233 and URL: <https://doi.org/10.5281/zenodo.15201233>). Additionally, all simulation results can be interactively reviewed through a Shiny web application at <https://ehsanx.shinyapps.io/hdPS-Alternatives/>.

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April 12, 2025

Cover Letter – Resubmission of PONE-D-25-00468

Title: *Is There a Competitive Advantage to Using Multivariate Statistical or Machine Learning Methods Over the Bross Formula in the hdPS Framework for Bias and Variance Estimation?*

Dear Dr. Adineh,

We thank you and the reviewers for the thorough and constructive feedback on our manuscript (PONE-D-25-00468). We are pleased to submit a revised version of our article entitled “Is There a Competitive Advantage to Using Multivariate Statistical or Machine Learning Methods Over the Bross Formula in the hdPS Framework for Bias and Variance Estimation?” for reconsideration in *PLOS ONE*.

We have carefully addressed all comments from the reviewers and the journal’s editorial team. A detailed Response to Reviewers is included, outlining how each point has been addressed. Revisions have been made in both the main text and figures/tables, with changes marked in red in the tracked version of the manuscript.

Below, we address the journal’s additional requirements:

1. Manuscript formatting:
We have ensured that the manuscript complies with *PLOS ONE* formatting requirements, including file naming and style.
2. Code availability:
All author-generated code and analysis scripts are openly available at our GitHub repository: <https://github.com/ehsanx/hdPS-proxy-select-codes>.
A citable, stable version of the code is archived on Zenodo:
<https://doi.org/10.5281/zenodo.15201233>.
3. Financial disclosure – Role of funders:
This work was supported by MEK’s NSERC Discovery Grant (PG#: 20R01603) and Discovery Launch Supplement (PG#: 20R12709).
The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript.
4. Competing interests statement:
MEK is currently supported by grants from CIHR and MS Canada. MEK has previously received consulting fees from Biogen Inc. for consulting unrelated to this current work. MEK was also previously supported by the Michael Smith Foundation for Health Research Scholar award.
This does not alter our adherence to PLOS ONE policies on sharing data and materials.
5. Data availability:
The minimal dataset and supporting files required to replicate our findings are deposited



in a stable public repository on Zenodo:
DOI: 10.5281/zenodo.15201233

6. Ethics statement:

The ethics statement has been relocated to the Methods section, as requested. The study involves only secondary de-identified data and complies with the ethical guidelines outlined by UBC's Policy 89 and TCPS2.

We sincerely hope that the revisions we have made adequately address the feedback and improve the clarity, rigor, and relevance of our work. Thank you for the opportunity to revise our manuscript.

Please do not hesitate to contact us if further clarifications are needed.

Sincerely,

A handwritten signature in blue ink that reads 'M Ehsan Karim'.

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Is There a Competitive Advantage to Using Multivariate Statistical or Machine Learning Methods Over the Bross Formula in the hdPS Framework for Bias and Variance Estimation?

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Abstract

Purpose: We aim to evaluate various proxy selection methods within the context of high-dimensional propensity score (hdPS) analysis. This study aimed to systematically evaluate and compare the performance of traditional statistical methods and machine learning approaches within the hdPS framework, focusing on key metrics such as bias, standard error (SE), and coverage, under various exposure and outcome prevalence scenarios. **Methods:** We conducted a plasmode simulation study using data from the National Health and Nutrition Examination Survey (NHANES) cycles from 2013 to 2018. We compared methods including the kitchen sink model, Bross-based hdPS, Hybrid hdPS, LASSO, Elastic Net, Random Forest, XGBoost, and Genetic Algorithm (GA). The performance of each inverse probability weighted method was assessed based on bias, MSE, coverage probability, and SE estimation across three epidemiological scenarios: frequent exposure and outcome, rare exposure and frequent outcome, and frequent exposure and rare outcome. **Results:** XGBoost consistently demonstrated strong performance in terms of MSE and coverage, making it effective for scenarios prioritizing precision. However, it exhibited higher bias, particularly in rare exposure scenarios, suggesting it is less suited when minimizing bias is critical. In contrast, GA showed significant limitations, with consistently high bias and MSE, making it the least reliable method. Bross-based hdPS, and Hybrid hdPS methods provided a balanced approach, with low bias and moderate MSE, though coverage varied depending on the scenario. Rare outcome scenarios generally resulted in lower MSE and better precision, while rare exposure scenarios were associated with higher bias and MSE. Notably, traditional statistical approaches such as forward selection and backward elimination performed comparably to more sophisticated machine learning methods in terms of bias and coverage, suggesting that these simpler approaches may be viable alternatives due to their computational efficiency. **Conclusion:** The results highlight the importance of selecting hdPS methods based on the specific characteristics of the data, such as exposure and outcome prevalence. While advanced machine learning methods such as XGBoost can enhance precision, simpler methods such as forward selection or backward elimination may offer similar performance in terms of bias and coverage with fewer computational demands. Tailoring the choice of method to the epidemiological scenario is essential for optimizing the balance between bias reduction and precision.

Background

High-dimensional Propensity Score (hdPS) Algorithm: Propensity scores are a crucial tool in causal inference for addressing confounding. A propensity score is the conditional probability of receiving a particular treatment or exposure, given a set of observed covariates. Traditional propensity score models rely on “investigator-specified” or manually selected covariates based on domain knowledge, prior literature, or theoretical frameworks.

In epidemiology, proxy variables are commonly used as substitutes for confounders that are difficult or impossible to measure directly, such as socioeconomic status, lifestyle factors, or health behaviors [1]. These variables are not directly identified as confounders but serve as surrogates for unmeasured or poorly measured confounders. The hdPS is a pharmacoepidemiological method designed to reduce confounding bias in large healthcare databases [2]. HdPS automatically ranks a wide array of proxy variables from healthcare records—such as diagnosis codes, medications, and procedures—using the Bross formula [3,4]. The Bross formula ranks these variables based on their marginal associations with both exposure and outcome. These selected proxy variables serve as surrogates for unmeasured or poorly measured confounders, helping reduce bias in treatment effect estimates. The hdPS algorithm further refines the selection by prioritizing variables based on their prevalence and potential for confounding, defined by their association with both exposure and outcome [2].

Importantly, hdPS has become a foundational method in modern epidemiologic research that uses healthcare databases, offering a scalable and reproducible approach to confounding control when traditional investigator-specified propensity score models are limited by unmeasured, mismeasured, or omitted variables. Unlike conventional PS methods that rely solely on expert knowledge and a fixed set of covariates, hdPS empirically identifies and prioritizes proxy variables from high-dimensional data, allowing it to adjust for latent confounding structures more effectively. This makes hdPS particularly valuable for improving causal inference in observational healthcare studies, including those evaluating drug safety, treatment effectiveness, and health system interventions [5,6].

Multivariate Machine Learning Extensions: Although the Bross formula performs well in certain contexts, it has limitations in capturing complex interactions, nonlinearities, and higher-order associations between variables, especially in high-dimensional settings where it does not account for the multivariate structure of other covariates [5,7]. To address these model-specification-related limitations, multivariate machine learning methods such as LASSO, Elastic Net, and Random Forests have been applied within the hdPS framework. These methods are better suited for high-dimensional data, where they can more effectively handle complex relationships and improve the selection of proxy variables, thus enhancing the precision of treatment effect estimates [7–9]. Several studies have proposed hybrid approaches that combine machine learning and hdPS to enhance variable selection and bias reduction while maintaining interpretability [6,7,9]. Simulation studies and empirical research have shown that these machine learning-based methods, or hybrid approaches combining the Bross formula with machine learning, can reduce confounding more effectively and increase efficiency compared to the Bross formula alone in certain settings [7–9]. These efforts reflect a growing interest in refining hdPS with modern statistical learning tools to optimize proxy confounder adjustment in realistic, data-rich settings.

Assessing the Simulation Performance: Other than bias, previous studies have primarily focused on Mean Squared Error (MSE) as a key metric for evaluating the performance of hdPS and its machine learning extensions [7,9–12]. However, in high-dimensional settings with singly robust methods, such as hdPS and machine learning approaches such as LASSO, MSE may not always be the most reliable measure.

MSE combines both bias and variance into a single metric, which makes interpretation challenging when variance estimation is unstable—a common issue in these methods. In contrast, coverage, which measures the proportion of confidence intervals that capture the true treatment effect, provides a more direct and meaningful assessment of a model’s reliability. In realistic observational studies, where model misspecification is often inevitable, coverage—along with related metrics such as bias-eliminated coverage and relative error in standard error (SE; which compares model-based SE with empirical SE)—can reveal whether confidence intervals or SEs are too narrow (underestimating uncertainty) or too wide (overestimating uncertainty). This insight is crucial in determining whether the model delivers valid estimates despite misspecification. Even a model with poor MSE but good coverage may still be valuable, as it produces realistic confidence intervals. By shifting the focus to coverage, rather than relying solely on MSE, we can achieve a more comprehensive understanding of method performance, especially in cases where unstable variance estimation might distort conclusions drawn from MSE alone. To extend prior research, our study systematically compares multiple machine learning and hybrid hdPS approaches using plasmode simulations grounded in real-world data structure. Importantly, we expand beyond commonly reported metrics such as bias and MSE by incorporating underutilized diagnostics—namely, empirical coverage and relative error in standard error—to provide a more nuanced understanding of estimator performance.

Aim: This research aims to systematically evaluate and compare various proxy selection methods within the hdPS framework for inverse probability weighted estimators, using a diverse range of simulation performance metrics, including bias, MSE, and coverage. The study focuses on assessing how these alternative statistical and machine learning methods perform in selecting proxy variables for confounding adjustment, compared to the traditional Bross formula.

Methods

Data and Simulation

Motivating Example: We revisited the association between obesity and the risk of diabetes using data from three cycles of the National Health and Nutrition Examination Survey (NHANES) covering the years 2013-2014, 2015-2016, and 2017-2018 [5]. To identify relevant investigator-specified covariates, we constructed a causal diagram based on literature [13–16] and established causal inference principles [17]. The covariates included in our analysis were carefully selected and categorized into demographic, behavioral, health history, access-related, and laboratory variables. While most of these variables were binary or categorical, the Laboratory variables were continuous. The secondary, de-identified data used in this study were accessed from the NHANES database on 1/5/2024-31/8/2024. The authors did not have access to any information that could identify individual participants during or after data collection. NHANES data are fully anonymized and publicly available, ensuring that individual privacy is protected.

Plasmode simulation: To rigorously assess the performance of the methods under consideration, we employed a plasmode simulation framework, which is particularly well-suited for reflecting real-world data structures and complexities [18]. This approach was inspired by the analytic dataset derived from NHANES and involved resampling from the observed covariates and exposure information (i.e., obesity) without altering them. By mirroring key aspects of an actual epidemiological study, this simulation framework offers a substantial advantage over traditional Monte Carlo simulations, which often rely on hypothetical assumptions.

Simulation scenarios under consideration: Our plasmode simulation was conducted over 500 iterations. For the base simulation scenario, we set the prevalence of exposure (obesity) and the event rate (diabetes) at 30%, with a true odds ratio (OR) parameter of 1, corresponding to a risk difference (RD) of 0. Each simulated dataset had a sample size of 3,000 participants. The description of other scenarios under consideration is provided in Table 1.

Table 1. Overview of Plasmode Simulation Scenarios Reflecting Varying Exposure and Outcome Prevalences Based on National Health and Nutrition Examination Survey (NHANES) Data Cycles (2013-2018)

Plasmode Simula- tion Scenario	Exposure Prevalence	Outcome Prevalence	True Odds Ratio	Sample Size
(i) Frequent Expo- sure and Outcome (Base)	30%	30%	1	3,000
(ii) Rare Exposure and Frequent Out- come	5%	30%	1	3,000
(iii) Frequent Expo- sure and Rare Out- come	30%	5%	1	3,000

True Data Generating Mechanism Used in Plasmode Simulation: The primary goal of this plasmode simulation study is to evaluate various variable selection methods under realistic conditions. To achieve this, we formulated the outcome data based on a specific model specification that incorporates both exposure and covariates, including investigator-specified and proxy variables. The model specification consists of three key components (See Appendices §A and B for further details):

- Investigator-Specified Covariates (demographic, behaviour and health history / access variables):* We retained the original investigator-specified covariates, which were either binary or categorical, reflecting how real-world studies typically operate. These covariates were not included in the pool of proxy variables or subjected to any variable selection or ranking processes. Instead, they were always included in the analysis as a separate set of variables based on prior knowledge and causal reasoning. This ensured that these known confounders were appropriately accounted for, independent of the proxy variable selection framework.
- Investigator-Specified Covariates (Transformation of Laboratory Variables):* In real-world studies, it is common for analysts to lack precise knowledge of the true model specification. To simulate this uncertainty, we transformed the continuous laboratory variables using complex functions such as logarithmic, exponential, square root, polynomial transformations, and interactions. This reflects the challenges analysts face in correctly specifying models when dealing with continuous data.
- Inclusion of Proxy Variables:* Real-world studies often deal with unmeasured confounding, which researchers attempt to mitigate by adding proxy variables. However, when a large number of proxies are added, some may act as noise variables, contributing little or nothing to the analysis. To simulate this, we selected only those binary proxy covariates (referred to as recurrence covariates in hdPS terminology) that had a relative risk (RR) of less than 0.8 or greater than

1.2 concerning the outcome. Proxy variables were defined based on their association with the outcome because unmeasured confounders must influence the outcome to induce confounding. By prioritizing variables with strong outcome associations, we aimed to ensure that the proxies reflect at least part of the confounding structure. While not explicitly enforced, many of the selected proxy variables are expected to have some degree of association with the exposure.

Out of 142 proxy covariates, 94 met this criterion and were summed in calculating a simple comorbidity burden measure [19]. This score represents the individual’s overall comorbidity burden, reflecting the cumulative presence of conditions. The remaining 48 covariates were excluded from this calculation and considered noise. This comorbidity burden measure (one variable) was then incorporated into our model specification for generating the outcome data.

Performance Measures: From this simulation, we derived several performance metrics to evaluate the effectiveness of the methods under consideration: (1) bias, (2) average model-based SE (the average of estimated SEs obtained from a model over repeated samples), (3) empirical SE (the standard deviation of estimated treatment effects across repeated samples), (4) MSE, (5) coverage probability of 95% confidence intervals, (6) bias-corrected coverage, and (7) Zip plot [20,21].

\textcolor{black}{Bias-corrected coverage provides a more accurate reflection of model reliability when bias is present. By adjusting for bias, we isolate the coverage performance attributable to variance estimation alone, allowing clearer comparisons of uncertainty calibration. This correction is crucial for distinguishing methods that produce narrow but biased intervals from those that offer valid inference under realistic misspecification.

While MSE is a commonly used metric, its interpretation becomes less reliable when variance estimates are unstable, as observed with singly robust machine learning models. In such cases, MSE may mask the trade-off between high variance and low bias (or vice versa), especially when confidence intervals are miscalibrated. Our inclusion of coverage and standard error diagnostics addresses this limitation by providing complementary insights into inference accuracy.

Inverse probability weighted estimators under consideration

The comparison between the data generation process and the analysis process reveals two key differences: (i) The data generation used transformed laboratory variables, whereas the analysis was conducted using only the original laboratory variables. (ii) The data generation employed a simple sum of selected proxy variables (sum of 94 proxy covariates), while the analysis included all proxy variables (142 binary proxies), with 48 of these acting as noise variables. These differences help us assess how the proxy variable selection methods handle model misspecification and the presence of noise variables.

We estimated propensity scores using the following models separately. Propensity scores were then used to construct weights for each individual, defined as the inverse of the probability of their observed exposure status. The primary causal estimand of interest in this study is the average treatment effect (ATE), which represents the difference in expected outcomes between treated and untreated individuals if all had been exposed to either treatment condition.

The selected machine learning models were chosen for their demonstrated ability to handle high-dimensional data and perform feature selection effectively. For example, XGBoost and Random Forest are tree-based ensemble methods known for capturing non-linear relationships and interactions. Genetic Algorithm (GA) offers a stochastic, optimization-based approach to variable selection, allowing exploration of complex covariate spaces. These models align with the goals of hdPS by enabling automated,

multivariate proxy variable selection that may capture complex dependencies missed by univariate selection strategies like the Bross formula. Incorporating such models expands the flexibility and potential robustness of hdPS analyses in high-dimensional data settings.

1. **Kitchen sink model:** The kitchen sink model serves as a comprehensive baseline that includes all available covariates—both investigator-specified and proxy variables—without any variable selection. It is used to benchmark the performance of variable selection methods against a model that assumes complete adjustment [7].
2. **hdPS using Bross formula:** The Bross formula is a statistical method used to calculate the bias introduced by not adjusting for a covariate [4]. In hdPS analysis, this formula was originally applied to each proxy variable to measure and rank the potential bias if the covariate were not adjusted for. In our analysis, the 100 proxies with the highest bias rankings are selected for further modeling [2,3]. This threshold of 100 proxy variables has been used in the literature to balance model complexity and overfitting risk, while ensuring that a sufficient number of proxies are included to capture unmeasured confounding [22]. Our choice reflects a practical trade-off between dimensionality and computational feasibility.
3. **Least Absolute Shrinkage and Selection Operator (LASSO):** LASSO is a variable selection technique that limits the number of variables by adding a penalty term to the regression model. Cross-validation (CV) is used in LASSO to identify variables with non-zero coefficients in the best model by optimizing the penalty value [7–9].
4. **Hybrid of hdPS and LASSO:** Instead of relying solely on LASSO for variable selection, a hybrid approach combines the Bross formula and LASSO. First, proxy variables are selected using the hdPS algorithm (e.g., the top 100), and then LASSO is applied to further refine the selection [7,9].
5. **Elastic Net:** Elastic Net is an extension of LASSO that includes an additional penalty term to handle multicollinearity by grouping correlated features and selecting the most representative ones [7].
6. **Random Forest:** The Random Forest algorithm is an ensemble learning method that constructs multiple decision trees to perform classification [23]. It calculates the importance of each proxy variable based on the decrease in impurity or Gini importance, providing a ranking of the proxies. The top 100 variables from this ranking are manually selected for further modeling [8].
7. **XGBoost:** XGBoost is a gradient boosting algorithm used to optimize machine learning models [24]. It builds decision trees that make splits based on maximum impurity reduction, and it assigns an importance score to each proxy variable by calculating the mean decrease in impurity [25].
8. **Stepwise:** Stepwise selection is a commonly used variable selection technique that builds or reduces models by adding or removing predictors based on statistical criteria—in our case, adjusted R-squared. We implemented two versions: (a) Forward selection (FS) begins with a minimal model, typically including only investigator-selected covariates. It then adds one proxy variable at a time—the one that most improves model performance at each step. This process continues until no additional variable meaningfully improves the fit. (b) Backward elimination (BE) takes the opposite approach. It starts with a full model

containing all investigator-selected and proxy variables and removes the least informative variable at each step, based on its contribution to model fit. This continues until removing more variables no longer improves or begins to harm the model. These stepwise approaches are transparent, making them accessible alternatives for confounding adjustment in high-dimensional settings.

9. **Genetic algorithm:** Genetic algorithm is an evolutionary algorithm inspired by the theory of natural selection [26]. It operates by evolving offspring from a population of the fittest individuals over several generations, evaluating and selecting the best combination of features or variables that maximize prediction accuracy.

Ethics approval and consent to participate

The analysis conducted on secondary and de-identified data is exempt from research ethics approval requirements. Ethics for this study was covered by item 7.10.3 in University of British Columbia's Policy no. 89: Research and Other Studies Involving Human Subjects 19 and Article 2.2 in of the Tri-Council Policy Statement- Ethical Conduct for Research Involving Humans (TCPS2).

Results

The results for each method under the different scenarios are summarized below. See Figs 1 and 2 for an overview of the performance in terms of bias and coverage, respectively. All simulation results can be reviewed interactively through a Shiny web application available at <https://ehsanx.shinyapps.io/hdPS-Alternatives/>, providing a convenient platform for exploring the performance of each method across various scenarios.

(i) Frequent Exposure and Outcome (base) scenario:

1. *Bias:* Bross-based hdPS exhibited the smallest bias (-0.0001), and the kitchen sink model (0.0002) was the second. Genetic algorithm shows the highest bias (0.0287), indicating a substantial deviation from the true effect. Among the other methods, Hybrid hdPS (0.0016), and Elastic Net (0.0036) demonstrated low bias. XGBoost (0.0074) had slightly higher bias than Random Forest (0.0034).
2. *Coverage:* The coverage for most methods was high, with Hybrid hdPS, Forward Selection, Backward Elimination, LASSO, and Elastic Net achieving values around 98%, indicating well-calibrated confidence intervals. However, Genetic algorithm had noticeably lower coverage (83.8%), indicating that its confidence intervals might be too narrow or biased, potentially missing the true effect. After applying bias elimination (as there were substantial bias associated with this method), Genetic algorithm's coverage improved to 96%.
3. *MSE:* XGBoost achieved the lowest MSE (0.0006). Genetic algorithm maintained the highest MSE (0.0016), reflecting its higher bias and variability. The kitchen sink model (0.0009), Bross-based hdPS (0.0008), Hybrid hdPS (0.0008), and Elastic Net (0.0009) all had relatively similar and moderate MSE values.
4. *SE:* XGBoost exhibited the lowest Empirical SE (0.0229), indicating high precision in its estimates. The kitchen sink model had the highest Empirical SE (0.0305), suggesting greater variability. Other methods, including Genetic algorithm (0.0274), Hybrid hdPS (0.0278), and Bross-based hdPS (0.0287), showed moderate

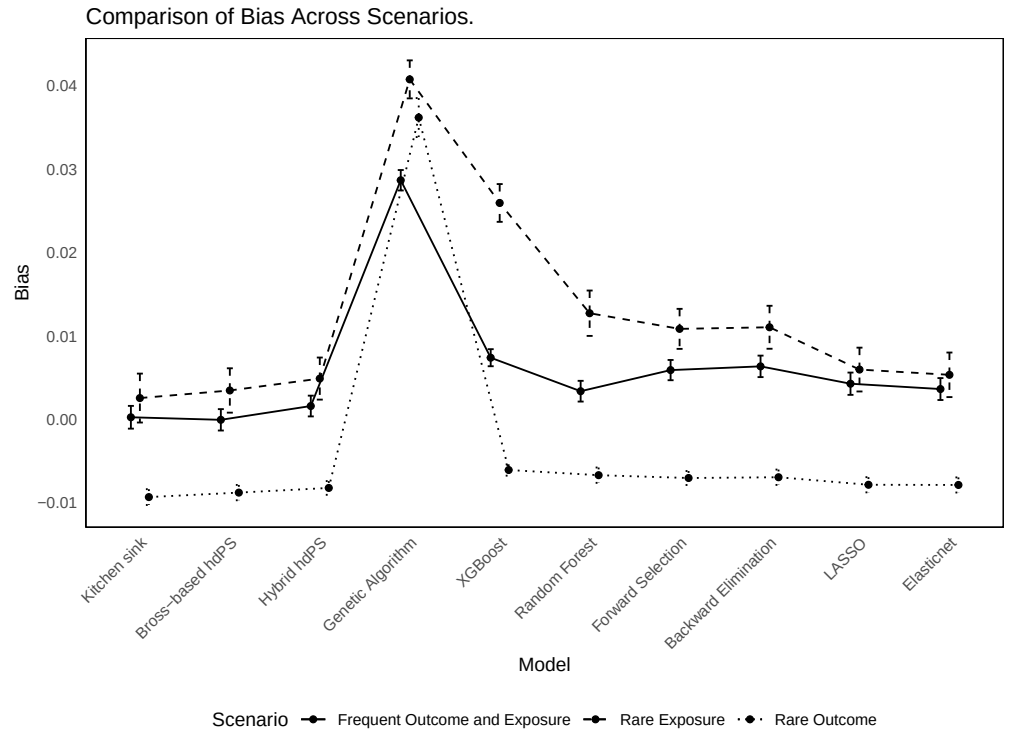


Fig 1. Comparison of Bias Across Different Methods in hdPS Analysis

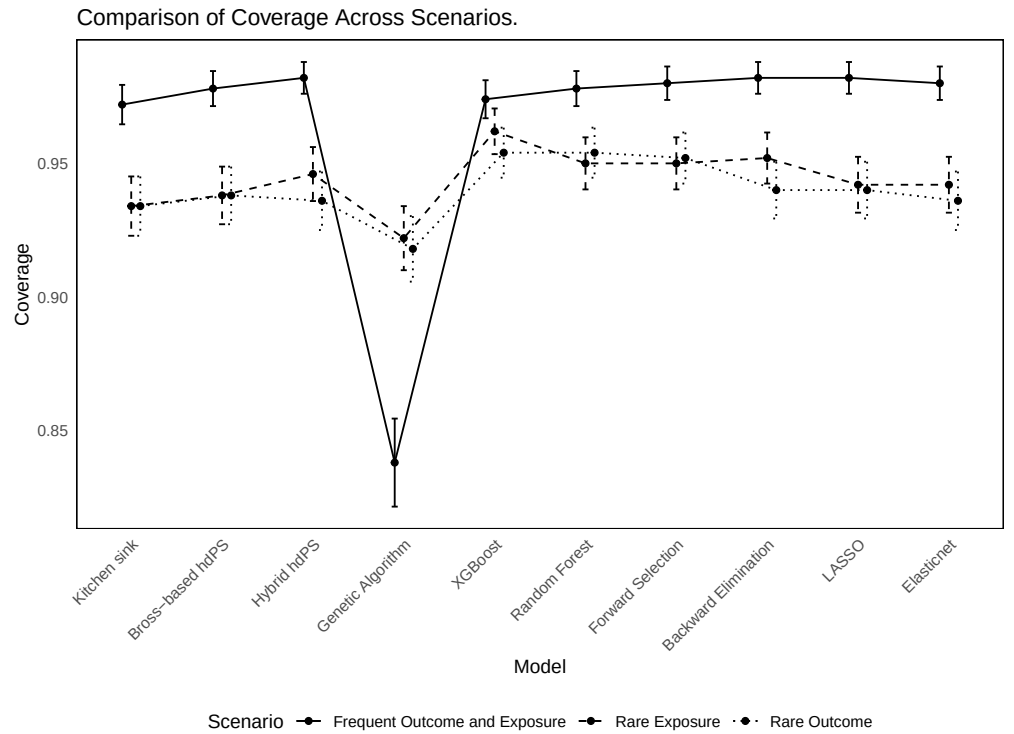


Fig 2. Comparison of Coverage Probability Across Different Methods in hdPS Analysis

variability. LASSO (0.0299) and Elastic Net (0.0294) had slightly higher variability. The Model-based SE followed a similar pattern, with XGBoost (0.0268) showing the lowest variability and the kitchen sink model (0.0333) the highest, indicating less precision in its estimates. When comparing relative error in SE estimation, XGBoost performed the worst. See Appendix §C for further details.

(ii) Rare Exposure and Frequent Outcome:

1. *Bias*: The kitchen sink model showed a relatively low bias (0.0025), while Genetic algorithm continued to exhibit the highest bias (0.0408), indicating a significant deviation from the true effect. XGBoost had a bias of 0.0259, which, while still higher than other methods, but was lower than Genetic algorithm. Other methods, such as Bross-based hdPS (0.0035), Hybrid hdPS (0.0049), and Elastic Net (0.0053), demonstrated moderate bias. Random Forest (0.0127) and Forward Selection (0.0108) had slightly higher bias but remained within an acceptable range.
2. *Coverage*: Coverage levels remained high for most methods, with XGBoost achieving the highest coverage at 96.2%, indicating well-calibrated confidence intervals despite its higher bias. The Genetic algorithm method had lower coverage (92.2%), suggesting that its confidence intervals might be narrower, potentially missing the true effect. Other methods such as RF, Forward Selection, Backward Elimination, and Hybrid hdPS maintained coverage values around 94-95%, suggesting adequate interval calibration. Bias-eliminated coverage for Genetic algorithm improved to 94.2%, but it was still slightly lower than other methods (e.g., forward selection).
3. *MSE*: Forward selection demonstrated the lowest MSE (0.0030), and then Hybrid hdPS and XGBoost. The Genetic algorithm method had a higher MSE (0.0043), reflecting its substantial bias and variability. The kitchen sink model also had an MSE of 0.0043, similar to Genetic algorithm, while other methods such as Bross-based hdPS (0.0035), RF (0.0039), and Elastic Net (0.0036) exhibited moderate MSE values, indicating reasonable accuracy.
4. *SE*: The lowest Empirical SE was observed with XGBoost (0.0507) and Genetic algorithm (0.0510), reflecting high precision despite their higher bias. The kitchen sink model had the highest Empirical SE (0.0656), indicating greater variability. Hybrid hdPS (0.0564), Bross-based hdPS (0.0595), and RF (0.0609) showed moderate variability. Forward Selection (0.0537) and Backward Elimination (0.0576) had lower variability compared to the kitchen sink model. In terms of Model-based SE, XGBoost (0.0531) and Genetic algorithm (0.0533) continued to show low variability, while the kitchen sink model had the highest Model-based SE (0.0623), indicating less precision in its estimates. When comparing relative error in SE estimation, XGBoost and kitchen sink model performed the worst (in other direction).

(iii) Frequent Exposure and Rare Outcome:

1. *Bias*: In this scenario, the kitchen sink model exhibited a moderate negative bias (-0.0093), similar to the Bross-based hdPS method (-0.0088). Genetic algorithm showed a significantly higher bias (0.0362), indicating a substantial deviation from the true effect. Among other methods, XGBoost demonstrated the lowest bias (-0.0061), while methods such as Hybrid hdPS (-0.0082), Forward Selection (-0.0070), and Backward Elimination (-0.0070) had slightly higher but still moderate biases. Elastic Net and LASSO both had biases of -0.0079, reflecting slightly larger deviations compared to XGBoost but still within acceptable limits.

2. *Coverage*: Most methods achieved good coverage, with XGBoost, RF, and Forward Selection each achieving a coverage rate of 95.4%, indicating well-calibrated confidence intervals. The Genetic algorithm method, however, had slightly lower coverage (91.8%), indicating that its confidence intervals might be narrower, potentially excluding the true effect. Bross-based hdPS and the kitchen sink model had slightly lower coverage values of 93.8% and 93.4%, respectively. After accounting for bias, the bias-eliminated coverage for most methods, except Genetic algorithm, remained high, with values ranging from 98.4% to 99.0%, indicating that most methods effectively adjusted for bias in their coverage estimates. Genetic algorithm's bias-eliminated coverage was lower at 93.4%, reflecting its higher inherent bias.
3. *MSE*: XGBoost exhibited the lowest MSE (0.0003). Genetic algorithm had the highest MSE (0.0040), reflecting its substantial bias and variability. The kitchen sink model (0.0005), Bross-based hdPS (0.0005), and other methods such as Hybrid hdPS (0.0004) and Elastic Net (0.0005) all had relatively similar MSE values, indicating moderate accuracy.
4. *SE*: The lowest Empirical SE was observed with XGBoost (0.0152), reflecting high precision in its estimates. The Genetic algorithm method exhibited the highest Empirical SE (0.0523), indicating greater variability and less precision. Methods such as Hybrid hdPS (0.0184), Forward Selection (0.0187), and Elastic Net (0.0203) showed moderate variability, while Bross-based hdPS (0.0206) and the kitchen sink model (0.0212) had slightly higher variability. In terms of Model-based SE, XGBoost (0.0179) again showed the lowest variability, consistent with its low Empirical SE, indicating that it provided the most stable estimates. The kitchen sink model had a slightly higher Model-based SE (0.0219), indicating less precision in its estimates. When comparing relative error in SE estimation, XGBoost performed the worst.

Real-world analysis

Summary results: The dataset comprises 7,585 individuals. Among these, the prevalence of the exposure is 48.8%, while the prevalence of the outcome is 23.7%.

See Fig 3 for the results from analyzing the NHANES (2013-2018) dataset. The methods are arranged according to the number of selected proxy variables. Among all variable selection algorithms, Forward Selection and LASSO yield the highest ORs, with values of 1.56 and 1.55, respectively. The ORs for the remaining methods cluster around 1.50 to 1.54. Additionally, with the exception of a few methods such as Kitchen Sink and Genetic Algorithm, a general pattern emerges in which methods selecting a smaller number of proxy variables tend to produce higher ORs. For RD, Forward Selection again shows the highest value at 0.082, followed closely by LASSO (0.080). Most other methods yield RD values between 0.075 and 0.079, reflecting a similar pattern to that observed for the ORs. This consistency suggests that the relationship between variable selection method and effect size persists across different estimands.

Table 2 presents a pairwise comparison of the number of proxy features shared between different variable selection methods used in the analysis. Each cell in the table indicates the count of common proxy variables selected by the method in the corresponding row and column. The diagonal cells, where the row and column methods are the same, represent the total number of proxy variables selected exclusively by each method. The first column displays the number of proxy variables shared between each method and the kitchen sink (KS) model. Given that the KS model includes all proxy variables, the values in the first column are identical to those in the diagonal cells for

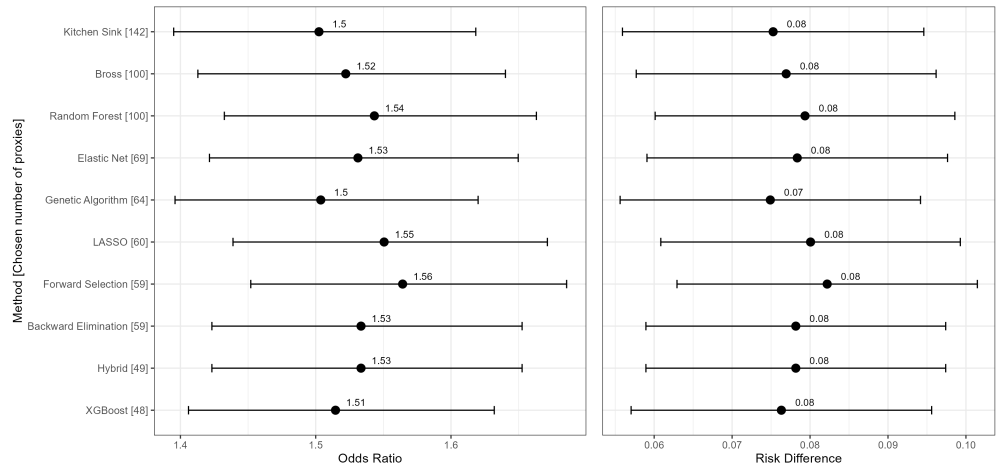


Fig 3. Figure presenting a comparison of Risk Differences (RD) and Odds Ratios (OR) with 95% confidence intervals for different methods used to evaluate the association between obesity and diabetes risk. The analysis is based on data from the National Health and Nutrition Examination Survey (NHANES) for the years 2013-2018. Methods are arranged by the number of variables used in the models.

each row, which also presents the total number of proxy variables selected by the method in the corresponding row.

374

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Table 2. Comparison of variable overlap of selected proxies across different methods used to evaluate the association between obesity and diabetes from the National Health and Nutrition Examination Survey (NHANES) for the years 2013-2018. Diagonal entries show the total number of proxies selected by each method, and off-diagonal entries represent the count of shared variables between method pairs. Most methods share a moderate number of proxies (typically 50–60 percent of the smaller set), indicating partial agreement in variable selection. Higher overlap is observed between closely related methods (e.g., LASSO and Elastic Net, or Hybrid with Bross/LASSO), while methods like XGBoost and Genetic Algorithm show lower overlap with others, reflecting divergent selection behavior in high-dimensional settings.

	KS	Bross	Hybrid	LASSO	EN	RF	XGB	FS	BE	GA
Kitchen sink (KS)	142									
Bross formula	100	100								
Hybrid (Bross and LASSO)	49	49	49							
LASSO	60	47	47	60						
Elastic Net (EN)	69	54	48	60	69					
Random Forest (RF)	100	71	38	46	52	100				
XGBoost (XGB)	48	38	24	28	30	37	48			
Forward selection (FS)	59	45	41	51	54	45	25	59		
Backward elimination (BE)	59	45	41	51	54	45	25	59	59	
Genetic algorithm (GA)	64	44	28	36	40	49	25	35	35	64

Table 3 presents a comparison of the count and percentage of common proxy variables selected by different methods in comparison with the Bross formula hdPS. The first column shows the total number of proxy variables selected by each method, while the second column lists the number of common features selected by both the respective method and the Bross formula. The third column reports the percentage of common features out of the total number of features selected by each method.

We observe that, aside from the Bross formula hdPS and Random Forest models, where the number of proxies was manually set to 100, and the kitchen sink (KS) model, which includes all proxies by design, the number of proxy variables selected by other models ranges between 49 and 69. As expected, the hybrid method combining Bross and LASSO hdPS selects exactly 49 proxy variables, all of which are selected by both methods, resulting in a common feature rate of 1.00. For other models, the common feature percentage is generally clustered around 74%, with XGBoost showing the highest common percentage at 79%, while GA displays the lowest common percentage at 69%.

Table 3. Comparison of the count and percentage of proxy variables selected by each methods in common with that by the Bross formula-based high-dimensional propensity score to evaluate the association between obesity and diabetes from the National Health and Nutrition Examination Survey (NHANES) for the years 2013-2018. The Hybrid method (Bross + LASSO) shows perfect agreement with the Bross-based hdPS by design. Other methods demonstrate overlap rates ranging from 0.69 to 0.79, indicating moderate consistency in variable selection. XGBoost shows the highest overlap (79 percent) among the non-hybrid methods, while the Genetic Algorithm shows the lowest (69 percent), reflecting greater divergence in selected proxies. These overlap patterns highlight methodological differences in how each approach prioritizes covariates in high-dimensional settings.

Method	Total Count	Common Count	Rate in Common
Kitchen sink	142	-	-
Bross formula	100	-	-
Hybrid (Bross and LASSO)	49	49	1.00
LASSO	60	47	0.78
Elastic Net	69	54	0.78
Random Forest	100	71	0.71
XGBoost	48	38	0.79
Forward selection	59	45	0.76
Backward elimination	59	45	0.76
Genetic algorithm	64	44	0.69

Computing time:
 Fig 4 presents the computing time for each method. All methods, aside from RF and GA, exhibit relatively fast computing times. RF and GA are generally much slower.

Discussion

Summary of the simulation findings

Comparison of methods:

Across the three scenarios, XGBoost consistently achieved the lowest MSE and high coverage, making it one of the most reliable methods in terms of precision. However, it consistently exhibited some degree of bias, particularly when compared to methods such

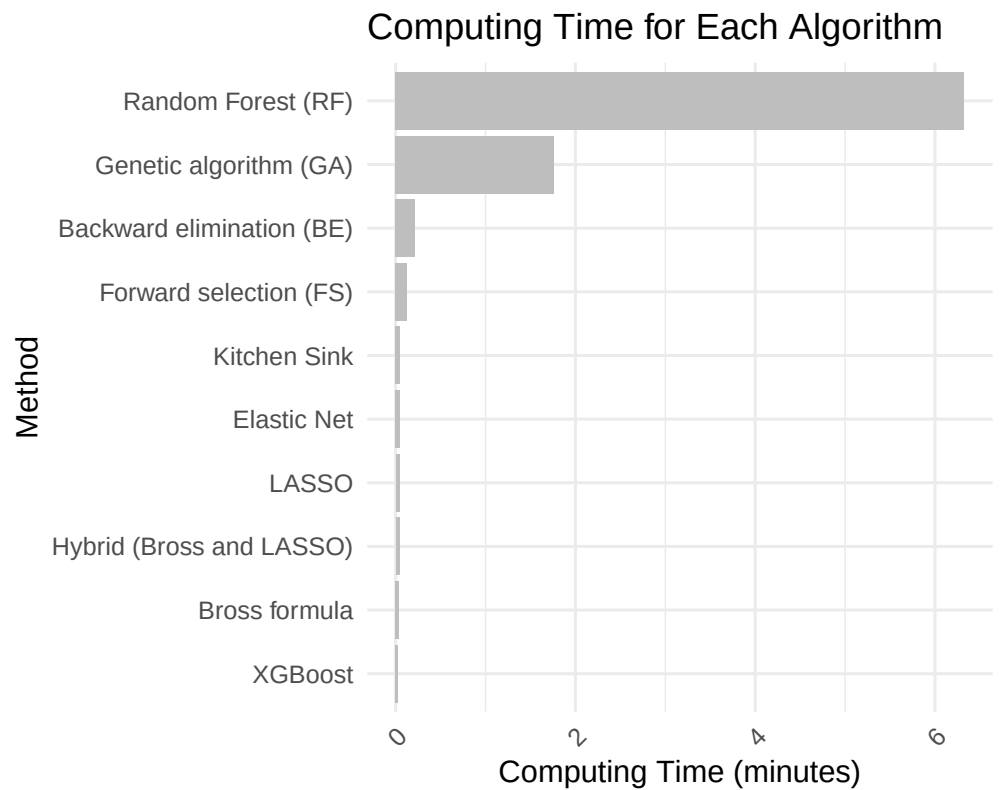


Fig 4. Computing time for the real-world analysis for each algorithm under consideration. The analysis is based on data from the National Health and Nutrition Examination Survey (NHANES) for the years 2013-2018.

as the kitchen sink model, Bross-based hdPS, and Hybrid hdPS, which often showed lower bias in scenarios with frequent outcomes. In contrast, GA displayed the highest bias and MSE, along with lower coverage and greater variability, making it the least reliable method for accurate effect estimation. Methods such as Bross-based hdPS, Hybrid hdPS, and Elastic Net performed moderately well across all scenarios, balancing bias, coverage, and MSE. However, these methods did not outperform XGBoost in terms of overall precision, especially with respect to MSE, though they often resulted in lower bias. The kitchen sink model performed comparably to Bross-based hdPS in terms of bias and coverage but lagged in SE estimation and MSE.

While XGBoost consistently yielded the lowest MSE across scenarios—indicating high overall accuracy—this gain came at the cost of increased bias, particularly in rare exposure settings. This pattern suggests that while XGBoost is highly efficient in minimizing prediction error, it may over-prioritize variance reduction at the expense of bias, potentially due to aggressive regularization or suboptimal feature weighting in the context of non-linear, sparse exposure structures. In causal inference applications, such trade-offs warrant caution, as models with high precision but biased estimates can result in systematically misleading conclusions regarding treatment effects.

GA exhibited consistently high bias and MSE, indicating substantial limitations in handling the dimensionality and noise characteristics inherent in the hdPS framework. One plausible explanation is that GA’s stochastic search process lacks the structure to efficiently navigate large covariate spaces with many weakly informative or correlated variables. Unlike regularized regression or tree-based methods, GA does not incorporate intrinsic mechanisms for managing multicollinearity or prioritizing marginally predictive features, which may lead to unstable or suboptimal variable subsets and subsequently poor confounding adjustment. Its computational intensity further compounds these limitations, making it less practical for high-dimensional epidemiologic analyses.

Comparison of scenarios:

In scenarios with rare exposure, higher bias was observed, particularly for methods such as GA and XGBoost, whereas frequent outcomes generally led to lower bias across most methods. Overcoverage was more common in the scenario with frequent exposure and outcome, with several methods producing confidence intervals that were too wide, suggesting an overestimation of uncertainty. The other scenarios exhibited more balanced or slightly under-coverage. Scenarios with frequent exposure also displayed higher relative error in SE estimation, making it more difficult to precisely estimate effects. In contrast, rare exposure scenarios were associated with higher MSE, reflecting the challenge of estimating effects when the exposure is uncommon. Rare outcome scenarios exhibited the lowest MSE across most methods (except GA), indicating that these scenarios provided a better balance between bias and precision for most methods.

These findings have practical implications for selecting variable selection strategies in epidemiological research. For studies focused on rare exposures (e.g., occupational or genetic risk factors), methods that minimize bias—such as Bross-based hdPS, Hybrid hdPS, or forward/backward stepwise selection—may be more suitable, as they provide more stable effect estimates under sparse data conditions. In contrast, studies targeting common conditions or aiming to improve precision, such as evaluations of widely used treatments or lifestyle exposures, may benefit from machine learning method such as XGBoost, which consistently demonstrate lower MSE. However, trade-offs between bias and variance should be carefully considered in relation to the study’s primary inference goal.

Contextualizing the literature

Previous studies have shown that LASSO performs well within the hdPS framework, particularly in terms of bias reduction and MSE [7,9], with similar results for Elastic

Net [7]. Our findings largely support these conclusions, as both methods demonstrated moderate bias and MSE across the scenarios. However, their performance was inconsistent in some cases, with higher bias observed in rare exposure scenarios, as noted in previous studies [7,9]. MSE also increased in rare exposure scenarios. The Hybrid hdPS method, combining Bross-based hdPS and LASSO, showed promising results, especially in terms of MSE, suggesting it may serve as a suitable alternative to traditional hdPS methods. Random Forest also performed similarly in our study, yielding relatively low bias, which is consistent with previous findings [7]. However, none of the earlier works emphasized coverage or examined methods such as GA, XGBoost, forward selection, or backward elimination, which were evaluated here.

Data analysis findings

The real-world dataset, with frequent exposure (48.8%) and a moderate outcome rate (23.7%), produced ORs of 1.56 and 1.55 for Forward Selection and LASSO, respectively, while other methods clustered around OR values between 1.50 and 1.54. A general trend emerged, showing that methods selecting fewer proxy variables tended to yield higher ORs, with exceptions such as the Kitchen Sink and Genetic Algorithm, which selected more proxies but produced lower ORs. In terms of risk difference (RD), estimates were relatively stable across methods, with Forward Selection again yielding the highest RD (0.082), followed by LASSO (0.080), while others ranged from 0.075 to 0.079. Regarding variable overlap, most methods shared around 74% of their proxy variables with the Bross formula hdPS, with XGBoost showing the highest common rate (79%) and GA the lowest (69%). Computing time analysis showed that, aside from RF and GA, most methods had relatively fast computing times, with RF being significantly slower.

In applied research, particularly with large-scale electronic health records or claims data, computational efficiency is a critical consideration. While tree-based or stochastic optimization methods may offer performance gains, their longer runtime may be impractical in settings requiring rapid analyses or model updates. Simpler approaches such as stepwise selection or penalized regression offer viable alternatives when speed, transparency, or reproducibility are prioritized.

Strengths

Previous studies on singly robust methods have mainly focused on hdPS performance using MSE as the primary evaluation metric [7,9–12]. Our study extends this body of work by incorporating a broader range of performance metrics, allowing researchers to compare results in terms of both bias and variance estimation. This comprehensive comparison of statistical and machine learning methods across various scenarios has not been conducted before. For instance, we found that while XGBoost consistently demonstrated strong performance in terms of MSE and coverage, it did not always have the lowest bias. Additionally, we observed that traditional variable selection methods such as forward selection and backward elimination performed similarly to more sophisticated methods such as LASSO and Random Forest across scenarios, both in terms of bias and coverage.

Our study also employed a complex plasmode simulation framework, closely replicating real-world data conditions [18]. This framework not only accounted for model misspecification, where the true relationships between covariates and outcomes were unknown, but also introduced noise variables, addressing the challenge of dealing with irrelevant covariates in high-dimensional settings. By applying transformations to continuous variables and adding proxy variables that contributed minimally to the outcome, we rigorously evaluated how well each method handled both model

uncertainty and non-informative variables, further strengthening the real-world
relevance of our findings.

Future Direction

Double robust methods have demonstrated strong potential for achieving optimal
statistical performance in hdPS analyses [10,27,28]. In addition, single robust methods,
such as ensemble learners such as super learners, have shown promise in improving bias,
MSE, area under the curve (AUC), and covariate balance in other contexts [11,29,30].
Deep learning methods, particularly supervised architectures, offer significant potential
for improving propensity score estimation in high-dimensional settings by capturing
complex, non-linear relationships and performing well in scenarios with rare exposures,
while maintaining comparable bias and superior variance estimation compared to
traditional methods [31]. Despite their theoretical advantages, the complexity and
computational demands of these methods, especially in high-dimensional settings, have
limited their adoption by practitioners. On the other hand, the simpler singly robust
machine learning methods evaluated here have been applied in clinical research [32,33].

Future research should explore the application of double robust methods and super
learners in hdPS analyses, particularly for handling rare outcomes and exposures, where
the performance of traditional methods may be suboptimal. Additionally, investigating
the impact of different hyperparameters on machine learning methods such as XGBoost
could optimize their performance in hdPS analysis. Finally, future simulation studies
should focus on evaluating coverage and other metrics in epidemiological scenarios such
as time-varying exposures, multiple treatment settings, and data with substantial
measurement error, which would provide valuable insights into the generalizability of
our findings [34].

One avenue for improving bias reduction in machine learning models within hdPS
analysis is the incorporation of causal structure or prior knowledge into variable
selection. Methods such as targeted maximum likelihood estimation, double machine
learning, or feature engineering based on directed acyclic graphs can help align machine
learning-based selection with the underlying causal model. Additionally, tuning
hyperparameters to favor less aggressive regularization or ensemble weighting may
reduce over-smoothing and improve bias performance, particularly in low-prevalence
strata.

Hybrid machine learning models—such as combinations of domain-informed methods
(e.g., Bross or investigator-specified variables) with data-driven algorithms like LASSO
or Random Forest—offer a promising balance between interpretability and predictive
accuracy. Their application in health data science could enable more robust causal
inference while accommodating real-world data challenges such as missingness,
nonlinearity, and high dimensionality. Such models may help mitigate confounding bias
without discarding variables of known clinical relevance, improving both validity and
stakeholder trust in observational research findings.

Conclusion

In conclusion, this analysis highlights the importance of carefully selecting appropriate
methods for hdPS analysis based on the specific characteristics of the data, particularly
the prevalence of exposure and outcome. These findings also emphasize the need to
tailor method selection to the specific epidemiological scenario, ensuring that the chosen
method aligns with the study’s goals, whether minimizing bias or maximizing precision.

XGBoost consistently demonstrated strong performance in terms of MSE and
coverage, making it an effective choice when precision is prioritized. However, it did not
achieve the lowest bias, particularly in rare exposure scenarios, indicating that it may

be less suited when minimizing bias is the primary objective. In contrast, the GA exhibited significant limitations, with consistently high bias and MSE, making it less reliable for effect estimation.

The kitchen sink, Bross-based hdPS, and Hybrid hdPS methods provided a more balanced approach, offering low bias and moderate MSE, though coverage varied depending on the scenario. The analysis also revealed that rare outcomes were associated with lower MSE and better precision, while rare exposures presented challenges, yielding higher bias and MSE. Interestingly, traditional statistical approaches such as forward selection and backward elimination performed comparably to more sophisticated machine learning methods across many scenarios. This suggests that simpler approaches can still be viable, particularly in terms of bias and coverage, and might be preferred due to their computational efficiency.

In real-world healthcare settings—such as electronic health record-based cohort studies, claims data analysis, or comparative effectiveness research—these results offer concrete guidance on choosing hdPS-compatible variable selection strategies. For instance, institutions conducting routine surveillance or drug safety studies may benefit from hybrid or regularized regression approaches that balance scalability with confounding control. Meanwhile, traditional stepwise methods remain attractive for smaller studies or settings requiring transparent modeling. Overall, aligning method choice with data availability, computational resources, and inferential goals can improve the practical utility of hdPS analyses in healthcare.

List of abbreviations

- hdPS: High-dimensional Propensity Score
- NHANES: National Health and Nutrition Examination Survey
- OR: Odds Ratio
- RD: Risk Difference
- SE: Standard Error
- MSE: Mean Squared Error
- KS: Kitchen Sink
- LASSO: Least Absolute Shrinkage and Selection Operator
- EN: Elastic Net; a regularized regression method that combines LASSO and Ridge regression
- RF: Random Forest
- XGBoost: Extreme Gradient Boosting
- FS: Forward Selection
- BE: Backward Elimination
- GA: Genetic Algorithm
- CV: Cross-Validation
- RR: Relative Risk

Declarations

Acknowledgment

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Consent for publication

The National Health and Nutrition Examination Survey (NHANES), conducted by the U.S. Centers for Disease Control and Prevention (CDC), involves collecting data through direct physical examinations, laboratory testing, and interviews. The CDC already obtains consent from participants when collecting this data. When researchers use NHANES data for their studies, they are typically using de-identified, publicly available data. This means that the information cannot be linked back to individual participants, and therefore, additional consent from participants is not required for researchers to use this data.

Availability of data and materials

NHANES data is publicly accessible and can be retrieved from the NHANES website. The datasets generated and/or analyzed during the current study are available in the NHANES repository: <https://www.cdc.gov/nchs/nhanes/index.htm>. Analysis data and codes are provided in <https://github.com/ehsanx/hdPS-proxy-select-codes>, with a stable, citable release archived at Zenodo (DOI: 10.5281/zenodo.15201233 and URL: <https://doi.org/10.5281/zenodo.15201233>). Additionally, all simulation results can be interactively reviewed through a Shiny web application at <https://ehsanx.shinyapps.io/hdPS-Alternatives/>.

Clinical trial number

Not applicable.

Competing interests

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Authors' contributions

MEK: Conceptualization, Writing – Original Draft, Review & Editing YL: Formal Analysis, Review & Editing

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References

1. VanderWeele TJ. Principles of confounder selection. *European journal of epidemiology*. 2019;34: 211–219.
2. Schneeweiss S, Rassen JA, Glynn RJ, Avorn J, Mogun H, Brookhart MA. High-dimensional propensity score adjustment in studies of treatment effects using health care claims data. *Epidemiology (Cambridge, Mass)*. 2009;20: 512.
3. Wyss R, Fireman B, Rassen JA, Schneeweiss S. Erratum: High-dimensional propensity score adjustment in studies of treatment effects using health care claims data. *Epidemiology*. 2018;29: e63–e64.
4. Bross ID. Spurious effects from an extraneous variable. *Journal of chronic diseases*. 1966;19: 637–647.
5. Karim ME. High-dimensional propensity score and its machine learning extensions in residual confounding control. *The American Statistician*. 2024; 1–38.
6. Wyss R, Yanover C, El-Hay T, Bennett D, Platt RW, Zullo AR, et al. Machine learning for improving high-dimensional proxy confounder adjustment in healthcare database studies: An overview of the current literature. *Pharmacoepidemiology and Drug Safety*. 2022;31: 932–943.
7. Karim ME, Pang M, Platt RW. Can we train machine learning methods to outperform the high-dimensional propensity score algorithm? *Epidemiology*. 2018;29: 191–198.
8. Schneeweiss S, Eddings W, Glynn RJ, Patorno E, Rassen J, Franklin JM. Variable selection for confounding adjustment in high-dimensional covariate spaces when analyzing healthcare databases. *Epidemiology*. 2017;28: 237–248.
9. Franklin JM, Eddings W, Glynn RJ, Schneeweiss S. Regularized regression versus the high-dimensional propensity score for confounding adjustment in secondary database analyses. *American journal of epidemiology*. 2015;182: 651–659.
10. Pang M, Schuster T, Filion KB, Schnitzer ME, Eberg M, Platt RW. Effect estimation in point-exposure studies with binary outcomes and high-dimensional covariate data—a comparison of targeted maximum likelihood estimation and inverse probability of treatment weighting. *The international journal of biostatistics*. 2016;12.
11. Wyss R, Schneeweiss S, Van Der Laan M, Lendle SD, Ju C, Franklin JM. Using super learner prediction modeling to improve high-dimensional propensity score estimation. *Epidemiology*. 2018;29: 96–106.
12. Simon V, Vadel J. Evaluating the performance of high-dimensional propensity scores compared with standard propensity scores for comparing antihypertensive therapies in the CPRD GOLD database. *Cardiology and Therapy*. 2023;12: 393–408.

13. Saydah S, Bullard KM, Cheng Y, Ali MK, Gregg EW, Geiss L, et al. Trends in cardiovascular disease risk factors by obesity level in adults in the united states, NHANES 1999-2010. *Obesity*. 2014;22: 1888–1895. 651
14. Liu J, Hay J, Faught BE, et al. The association of sleep disorder, obesity status, and diabetes mellitus among US adults—the NHANES 2009-2010 survey results. *International journal of endocrinology*. 2013;2013. 652
15. Kabadi SM, Lee BK, Liu L. Joint effects of obesity and vitamin d insufficiency on insulin resistance and type 2 diabetes: Results from the NHANES 2001–2006. *Diabetes care*. 2012;35: 2048–2054. 653
16. Ostchega Y, Hughes JP, Terry A, Fakhouri TH, Miller I. Abdominal obesity, body mass index, and hypertension in US adults: NHANES 2007–2010. *American journal of hypertension*. 2012;25: 1271–1278. 654
17. Greenland S, Pearl J, Robins JM. Causal diagrams for epidemiologic research. *Epidemiology*. 1999; 37–48. 655
18. Franklin JM, Schneeweiss S, Polinski JM, Rassen JA. Plasmode simulation for the evaluation of pharmacoepidemiologic methods in complex healthcare databases. *Computational statistics and data analysis*. 2014;72: 219–226. 656
19. Farley JF, Harley CR, Devine JW. A comparison of comorbidity measurements to predict healthcare expenditures. *American Journal of Managed Care*. 2006;12: 110–118. 657
20. Morris TP, White IR, Crowther MJ. Using simulation studies to evaluate statistical methods. *Statistics in medicine*. 2019;38: 2074–2102. 658
21. White IR, Pham TM, Quartagno M, Morris TP. How to check a simulation study. *International Journal of Epidemiology*. 2023; dyad134. 659
22. Enders D, Ohlmeier C, Garbe E. The potential of high-dimensional propensity scores in health services research: An exemplary study on the quality of care for elective percutaneous coronary interventions. *Health services research*. 2018;53: 197–213. 660
23. Breiman L. Random forests. *Machine learning*. 2001;45: 5–32. 661
24. Chen T, Guestrin C. Xgboost: A scalable tree boosting system. *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining*. 2016. pp. 785–794. 662
25. Xiao Y, Chen Y, Huang R, Jiang F, Zhou J, Yang T. Interpretable machine learning in predicting drug-induced liver injury among tuberculosis patients: Model development and validation study. *BMC Medical Research Methodology*. 2024;24: 92. 663

26. Holland JH. Adaptation in natural and artificial systems. University of Michigan Press; 1975. 677
678
27. Pang M, Schuster T, Filion KB, Eberg M, Platt RW. Targeted maximum likelihood estimation for pharmacoepidemiologic research. *Epidemiology (Cambridge, Mass)*. 2016;27: 570. 679
680
28. Benasseur I, Talbot D, Durand M, Holbrook A, Matteau A, Potter BJ, et al. A comparison of confounder selection and adjustment methods for estimating causal effects using large healthcare databases. *Pharmacoepidemiology and Drug Safety*. 2022;31: 424–433. 681
682
29. Guertin JR, Rahme E, Dormuth CR, LeLorier J. Head to head comparison of the propensity score and the high-dimensional propensity score matching methods. *BMC medical research methodology*. 2016;16: 1–10. 683
684
30. Ju C, Combs M, Lendle SD, Franklin JM, Wyss R, Schneeweiss S, et al. Propensity score prediction for electronic healthcare databases using super learner and high-dimensional propensity score methods. *Journal of Applied Statistics*. 2019;46: 2216–2236. 685
686
31. Karim ME. Can supervised deep learning architecture outperform autoencoders in building propensity score models for matching? *BMC Medical Research Methodology*. 2024;24: 167. 687
688
32. Hossain MB, Johnston JC, Cook VJ, Sadatsafavi M, Wong H, Romanowski K, et al. Role of latent tuberculosis infection on elevated risk of cardiovascular disease: A population-based cohort study of immigrants in british columbia, canada, 1985–2019. *Epidemiology & Infection*. 2023;151: e68. 689
690
33. Basham CA, Karim ME, Cook VJ, Patrick DM, Johnston JC. Post-tuberculosis airway disease: A population-based cohort study of people immigrating to british columbia, canada, 1985–2015. *EClinicalMedicine*. 2021;33. 691
692
34. Neugebauer R, Schmittdiel JA, Zhu Z, Rassen JA, Seeger JD, Schneeweiss S. High-dimensional propensity score algorithm in comparative effectiveness research with time-varying interventions. *Statistics in medicine*. 2015;34: 753–781. 693
694

Is There a Competitive Advantage to Using Multivariate Statistical or Machine Learning Methods Over the Bross Formula in the hdPS Framework for Bias and Variance Estimation?

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Abstract

Purpose: We aim to evaluate various proxy selection methods within the context of high-dimensional propensity score (hdPS) analysis. This study aimed to systematically evaluate and compare the performance of traditional statistical methods and machine learning approaches within the hdPS framework, focusing on key metrics such as bias, standard error (SE), and coverage, under various exposure and outcome prevalence scenarios. **Methods:** We conducted a plasmode simulation study using data from the National Health and Nutrition Examination Survey (NHANES) cycles from 2013 to 2018. We compared methods including the kitchen sink model, Bross-based hdPS, Hybrid hdPS, LASSO, Elastic Net, Random Forest, XGBoost, and Genetic Algorithm (GA). The performance of each inverse probability weighted method was assessed based on bias, MSE, coverage probability, and SE estimation across three epidemiological scenarios: frequent exposure and outcome, rare exposure and frequent outcome, and frequent exposure and rare outcome. **Results:** XGBoost consistently demonstrated strong performance in terms of MSE and coverage, making it effective for scenarios prioritizing precision. However, it exhibited higher bias, particularly in rare exposure scenarios, suggesting it is less suited when minimizing bias is critical. In contrast, GA showed significant limitations, with consistently high bias and MSE, making it the least reliable method. Bross-based hdPS, and Hybrid hdPS methods provided a balanced approach, with low bias and moderate MSE, though coverage varied depending on the scenario. Rare outcome scenarios generally resulted in lower MSE and better precision, while rare exposure scenarios were associated with higher bias and MSE. Notably, traditional statistical approaches such as forward selection and backward elimination performed comparably to more sophisticated machine learning methods in terms of bias and coverage, suggesting that these simpler approaches may be viable alternatives due to their computational efficiency. **Conclusion:** The results highlight the importance of selecting hdPS methods based on the specific characteristics of the data, such as exposure and outcome prevalence. While advanced machine learning methods such as XGBoost can enhance precision, simpler methods such as forward selection or backward elimination may offer similar performance in terms of bias and coverage with fewer computational demands. Tailoring the choice of method to the epidemiological scenario is essential for optimizing the balance between bias reduction and precision.

Background

High-dimensional Propensity Score (hdPS) Algorithm: Propensity scores are a crucial tool in causal inference for addressing confounding. A propensity score is the conditional probability of receiving a particular treatment or exposure, given a set of observed covariates. Traditional propensity score models rely on “investigator-specified” or manually selected covariates based on domain knowledge, prior literature, or theoretical frameworks.

In epidemiology, proxy variables are commonly used as substitutes for confounders that are difficult or impossible to measure directly, such as socioeconomic status, lifestyle factors, or health behaviors [1]. These variables are not directly identified as confounders but serve as surrogates for unmeasured or poorly measured confounders. The hdPS is a pharmacoepidemiological method designed to reduce confounding bias in large healthcare databases [2]. HdPS automatically ranks a wide array of proxy variables from healthcare records—such as diagnosis codes, medications, and procedures—using the Bross formula [3,4]. The Bross formula ranks these variables based on their marginal associations with both exposure and outcome. These selected proxy variables serve as surrogates for unmeasured or poorly measured confounders, helping reduce bias in treatment effect estimates. The hdPS algorithm further refines the selection by prioritizing variables based on their prevalence and potential for confounding, defined by their association with both exposure and outcome [2].

Importantly, hdPS has become a foundational method in modern epidemiologic research that uses healthcare databases, offering a scalable and reproducible approach to confounding control when traditional investigator-specified propensity score models are limited by unmeasured, mismeasured, or omitted variables. Unlike conventional PS methods that rely solely on expert knowledge and a fixed set of covariates, hdPS empirically identifies and prioritizes proxy variables from high-dimensional data, allowing it to adjust for latent confounding structures more effectively. This makes hdPS particularly valuable for improving causal inference in observational healthcare studies, including those evaluating drug safety, treatment effectiveness, and health system interventions [5,6].

Multivariate Machine Learning Extensions: Although the Bross formula performs well in certain contexts, it has limitations in capturing complex interactions, nonlinearities, and higher-order associations between variables, especially in high-dimensional settings where it does not account for the multivariate structure of other covariates [5,7]. To address these model-specification-related limitations, multivariate machine learning methods such as LASSO, Elastic Net, and Random Forests have been applied within the hdPS framework. These methods are better suited for high-dimensional data, where they can more effectively handle complex relationships and improve the selection of proxy variables, thus enhancing the precision of treatment effect estimates [7–9]. Several studies have proposed hybrid approaches that combine machine learning and hdPS to enhance variable selection and bias reduction while maintaining interpretability [6,7,9]. Simulation studies and empirical research have shown that these machine learning-based methods, or hybrid approaches combining the Bross formula with machine learning, can reduce confounding more effectively and increase efficiency compared to the Bross formula alone in certain settings [7–9]. These efforts reflect a growing interest in refining hdPS with modern statistical learning tools to optimize proxy confounder adjustment in realistic, data-rich settings.

Assessing the Simulation Performance: Other than bias, previous studies have primarily focused on Mean Squared Error (MSE) as a key metric for evaluating the performance of hdPS and its machine learning extensions [7,9–12]. However, in high-dimensional settings with singly robust methods, such as hdPS and machine learning approaches such as LASSO, MSE may not always be the most reliable measure.

MSE combines both bias and variance into a single metric, which makes interpretation challenging when variance estimation is unstable—a common issue in these methods. In contrast, coverage, which measures the proportion of confidence intervals that capture the true treatment effect, provides a more direct and meaningful assessment of a model’s reliability. In realistic observational studies, where model misspecification is often inevitable, coverage—along with related metrics such as bias-eliminated coverage and relative error in standard error (SE; which compares model-based SE with empirical SE)—can reveal whether confidence intervals or SEs are too narrow (underestimating uncertainty) or too wide (overestimating uncertainty). This insight is crucial in determining whether the model delivers valid estimates despite misspecification. Even a model with poor MSE but good coverage may still be valuable, as it produces realistic confidence intervals. By shifting the focus to coverage, rather than relying solely on MSE, we can achieve a more comprehensive understanding of method performance, especially in cases where unstable variance estimation might distort conclusions drawn from MSE alone. **To extend prior research, our study systematically compares multiple machine learning and hybrid hdPS approaches using plasmode simulations grounded in real-world data structure. Importantly, we expand beyond commonly reported metrics such as bias and MSE by incorporating underutilized diagnostics—namely, empirical coverage and relative error in standard error—to provide a more nuanced understanding of estimator performance.**

Aim: This research aims to systematically evaluate and compare various proxy selection methods within the hdPS framework for inverse probability weighted estimators, using a diverse range of simulation performance metrics, including bias, MSE, and coverage. The study focuses on assessing how these alternative statistical and machine learning methods perform in selecting proxy variables for confounding adjustment, compared to the traditional Bross formula.

Methods

Data and Simulation

Motivating Example: We revisited the association between obesity and the risk of diabetes using data from three cycles of the National Health and Nutrition Examination Survey (NHANES) covering the years 2013-2014, 2015-2016, and 2017-2018 [5]. To identify relevant investigator-specified covariates, we constructed a causal diagram based on literature [13–16] and established causal inference principles [17]. The covariates included in our analysis were carefully selected and categorized into demographic, behavioral, health history, access-related, and laboratory variables. While most of these variables were binary or categorical, the Laboratory variables were continuous. The secondary, de-identified data used in this study were accessed from the NHANES database on 1/5/2024-31/8/2024. The authors did not have access to any information that could identify individual participants during or after data collection. NHANES data are fully anonymized and publicly available, ensuring that individual privacy is protected.

Plasmode simulation: To rigorously assess the performance of the methods under consideration, we employed a plasmode simulation framework, which is particularly well-suited for reflecting real-world data structures and complexities [18]. This approach was inspired by the analytic dataset derived from NHANES and involved resampling from the observed covariates and exposure information (i.e., obesity) without altering them. By mirroring key aspects of an actual epidemiological study, this simulation framework offers a substantial advantage over traditional Monte Carlo simulations, which often rely on hypothetical assumptions.

Simulation scenarios under consideration: Our plasmode simulation was conducted over 500 iterations. For the base simulation scenario, we set the prevalence of exposure (obesity) and the event rate (diabetes) at 30%, with a true odds ratio (OR) parameter of 1, corresponding to a risk difference (RD) of 0. Each simulated dataset had a sample size of 3,000 participants. The description of other scenarios under consideration is provided in Table 1.

Table 1. Overview of Plasmode Simulation Scenarios Reflecting Varying Exposure and Outcome Prevalences Based on National Health and Nutrition Examination Survey (NHANES) Data Cycles (2013-2018)

Plasmode Simula- tion Scenario	Exposure Prevalence	Outcome Prevalence	True Odds Ratio	Sample Size
(i) Frequent Expo- sure and Outcome (Base)	30%	30%	1	3,000
(ii) Rare Exposure and Frequent Out- come	5%	30%	1	3,000
(iii) Frequent Expo- sure and Rare Out- come	30%	5%	1	3,000

True Data Generating Mechanism Used in Plasmode Simulation: The primary goal of this plasmode simulation study is to evaluate various variable selection methods under realistic conditions. To achieve this, we formulated the outcome data based on a specific model specification that incorporates both exposure and covariates, including investigator-specified and proxy variables. The model specification consists of three key components (See Appendices §A and B for further details):

- Investigator-Specified Covariates (demographic, behaviour and health history / access variables):* We retained the original investigator-specified covariates, which were either binary or categorical, reflecting how real-world studies typically operate. These covariates were not included in the pool of proxy variables or subjected to any variable selection or ranking processes. Instead, they were always included in the analysis as a separate set of variables based on prior knowledge and causal reasoning. This ensured that these known confounders were appropriately accounted for, independent of the proxy variable selection framework.
- Investigator-Specified Covariates (Transformation of Laboratory Variables):* In real-world studies, it is common for analysts to lack precise knowledge of the true model specification. To simulate this uncertainty, we transformed the continuous laboratory variables using complex functions such as logarithmic, exponential, square root, polynomial transformations, and interactions. This reflects the challenges analysts face in correctly specifying models when dealing with continuous data.
- Inclusion of Proxy Variables:* Real-world studies often deal with unmeasured confounding, which researchers attempt to mitigate by adding proxy variables. However, when a large number of proxies are added, some may act as noise variables, contributing little or nothing to the analysis. To simulate this, we selected only those binary proxy covariates (referred to as recurrence covariates in hdPS terminology) that had a relative risk (RR) of less than 0.8 or greater than

1.2 concerning the outcome. Proxy variables were defined based on their association with the outcome because unmeasured confounders must influence the outcome to induce confounding. By prioritizing variables with strong outcome associations, we aimed to ensure that the proxies reflect at least part of the confounding structure. While not explicitly enforced, many of the selected proxy variables are expected to have some degree of association with the exposure.

Out of 142 proxy covariates, 94 met this criterion and were summed in calculating a simple comorbidity burden measure [19]. This score represents the individual's overall comorbidity burden, reflecting the cumulative presence of conditions. The remaining 48 covariates were excluded from this calculation and considered noise. This comorbidity burden measure (one variable) was then incorporated into our model specification for generating the outcome data.

Performance Measures: From this simulation, we derived several performance metrics to evaluate the effectiveness of the methods under consideration: (1) bias, (2) average model-based SE (the average of estimated SEs obtained from a model over repeated samples), (3) empirical SE (the standard deviation of estimated treatment effects across repeated samples), (4) MSE, (5) coverage probability of 95% confidence intervals, (6) bias-corrected coverage, and (7) Zip plot [20,21].

\textcolor{red}{Bias-corrected coverage provides a more accurate reflection of model reliability when bias is present. By adjusting for bias, we isolate the coverage performance attributable to variance estimation alone, allowing clearer comparisons of uncertainty calibration. This correction is crucial for distinguishing methods that produce narrow but biased intervals from those that offer valid inference under realistic misspecification.

\textcolor{red}{While MSE is a commonly used metric, its interpretation becomes less reliable when variance estimates are unstable, as observed with singly robust machine learning models. In such cases, MSE may mask the trade-off between high variance and low bias (or vice versa), especially when confidence intervals are miscalibrated. Our inclusion of coverage and standard error diagnostics addresses this limitation by providing complementary insights into inference accuracy.

Inverse probability weighted estimators under consideration

The comparison between the data generation process and the analysis process reveals two key differences: (i) The data generation used transformed laboratory variables, whereas the analysis was conducted using only the original laboratory variables. (ii) The data generation employed a simple sum of selected proxy variables (sum of 94 proxy covariates), while the analysis included all proxy variables (142 binary proxies), with 48 of these acting as noise variables. These differences help us assess how the proxy variable selection methods handle model misspecification and the presence of noise variables.

We estimated propensity scores using the following models separately. Propensity scores were then used to construct weights for each individual, defined as the inverse of the probability of their observed exposure status. The primary causal estimand of interest in this study is the average treatment effect (ATE), which represents the difference in expected outcomes between treated and untreated individuals if all had been exposed to either treatment condition.

\textcolor{red}{The selected machine learning models were chosen for their demonstrated ability to handle high-dimensional data and perform feature selection effectively. For example, XGBoost and Random Forest are tree-based ensemble methods known for capturing non-linear relationships and interactions. Genetic Algorithm (GA) offers a stochastic, optimization-based approach to variable selection, allowing exploration of complex covariate spaces. These models align with the goals of hdPS by enabling automated,

multivariate proxy variable selection that may capture complex dependencies missed by univariate selection strategies like the Bross formula. Incorporating such models expands the flexibility and potential robustness of hdPS analyses in high-dimensional data settings.

1. **Kitchen sink model:** The kitchen sink model serves as a comprehensive baseline that includes all available covariates—both investigator-specified and proxy variables—without any variable selection. It is used to benchmark the performance of variable selection methods against a model that assumes complete adjustment [7].
2. **hdPS using Bross formula:** The Bross formula is a statistical method used to calculate the bias introduced by not adjusting for a covariate [4]. In hdPS analysis, this formula was originally applied to each proxy variable to measure and rank the potential bias if the covariate were not adjusted for. In our analysis, the 100 proxies with the highest bias rankings are selected for further modeling [2,3]. This threshold of 100 proxy variables has been used in the literature to balance model complexity and overfitting risk, while ensuring that a sufficient number of proxies are included to capture unmeasured confounding [22]. Our choice reflects a practical trade-off between dimensionality and computational feasibility.
3. **Least Absolute Shrinkage and Selection Operator (LASSO):** LASSO is a variable selection technique that limits the number of variables by adding a penalty term to the regression model. Cross-validation (CV) is used in LASSO to identify variables with non-zero coefficients in the best model by optimizing the penalty value [7–9].
4. **Hybrid of hdPS and LASSO:** Instead of relying solely on LASSO for variable selection, a hybrid approach combines the Bross formula and LASSO. First, proxy variables are selected using the hdPS algorithm (e.g., the top 100), and then LASSO is applied to further refine the selection [7,9].
5. **Elastic Net:** Elastic Net is an extension of LASSO that includes an additional penalty term to handle multicollinearity by grouping correlated features and selecting the most representative ones [7].
6. **Random Forest:** The Random Forest algorithm is an ensemble learning method that constructs multiple decision trees to perform classification [23]. It calculates the importance of each proxy variable based on the decrease in impurity or Gini importance, providing a ranking of the proxies. The top 100 variables from this ranking are manually selected for further modeling [8].
7. **XGBoost:** XGBoost is a gradient boosting algorithm used to optimize machine learning models [24]. It builds decision trees that make splits based on maximum impurity reduction, and it assigns an importance score to each proxy variable by calculating the mean decrease in impurity [25].
8. **Stepwise:** Stepwise selection is a commonly used variable selection technique that builds or reduces models by adding or removing predictors based on statistical criteria—in our case, adjusted R-squared. We implemented two versions: (a) Forward selection (FS) begins with a minimal model, typically including only investigator-selected covariates. It then adds one proxy variable at a time—the one that most improves model performance at each step. This process continues until no additional variable meaningfully improves the fit. (b) Backward elimination (BE) takes the opposite approach. It starts with a full model

containing all investigator-selected and proxy variables and removes the least informative variable at each step, based on its contribution to model fit. This continues until removing more variables no longer improves or begins to harm the model. These stepwise approaches are transparent, making them accessible alternatives for confounding adjustment in high-dimensional settings.

9. **Genetic algorithm:** Genetic algorithm is an evolutionary algorithm inspired by the theory of natural selection [26]. It operates by evolving offspring from a population of the fittest individuals over several generations, evaluating and selecting the best combination of features or variables that maximize prediction accuracy.

Ethics approval and consent to participate

The analysis conducted on secondary and de-identified data is exempt from research ethics approval requirements. Ethics for this study was covered by item 7.10.3 in University of British Columbia's Policy no. 89: Research and Other Studies Involving Human Subjects 19 and Article 2.2 in of the Tri-Council Policy Statement- Ethical Conduct for Research Involving Humans (TCPS2).

Results

The results for each method under the different scenarios are summarized below. See Figs 1 and 2 for an overview of the performance in terms of bias and coverage, respectively. All simulation results can be reviewed interactively through a Shiny web application available at <https://ehsanx.shinyapps.io/hdPS-Alternatives/>, providing a convenient platform for exploring the performance of each method across various scenarios.

(i) Frequent Exposure and Outcome (base) scenario:

1. *Bias:* Bross-based hdPS exhibited the smallest bias (-0.0001), and the kitchen sink model (0.0002) was the second. Genetic algorithm shows the highest bias (0.0287), indicating a substantial deviation from the true effect. Among the other methods, Hybrid hdPS (0.0016), and Elastic Net (0.0036) demonstrated low bias. XGBoost (0.0074) had slightly higher bias than Random Forest (0.0034).
2. *Coverage:* The coverage for most methods was high, with Hybrid hdPS, Forward Selection, Backward Elimination, LASSO, and Elastic Net achieving values around 98%, indicating well-calibrated confidence intervals. However, Genetic algorithm had noticeably lower coverage (83.8%), indicating that its confidence intervals might be too narrow or biased, potentially missing the true effect. After applying bias elimination (as there were substantial bias associated with this method), Genetic algorithm's coverage improved to 96%.
3. *MSE:* XGBoost achieved the lowest MSE (0.0006). Genetic algorithm maintained the highest MSE (0.0016), reflecting its higher bias and variability. The kitchen sink model (0.0009), Bross-based hdPS (0.0008), Hybrid hdPS (0.0008), and Elastic Net (0.0009) all had relatively similar and moderate MSE values.
4. *SE:* XGBoost exhibited the lowest Empirical SE (0.0229), indicating high precision in its estimates. The kitchen sink model had the highest Empirical SE (0.0305), suggesting greater variability. Other methods, including Genetic algorithm (0.0274), Hybrid hdPS (0.0278), and Bross-based hdPS (0.0287), showed moderate

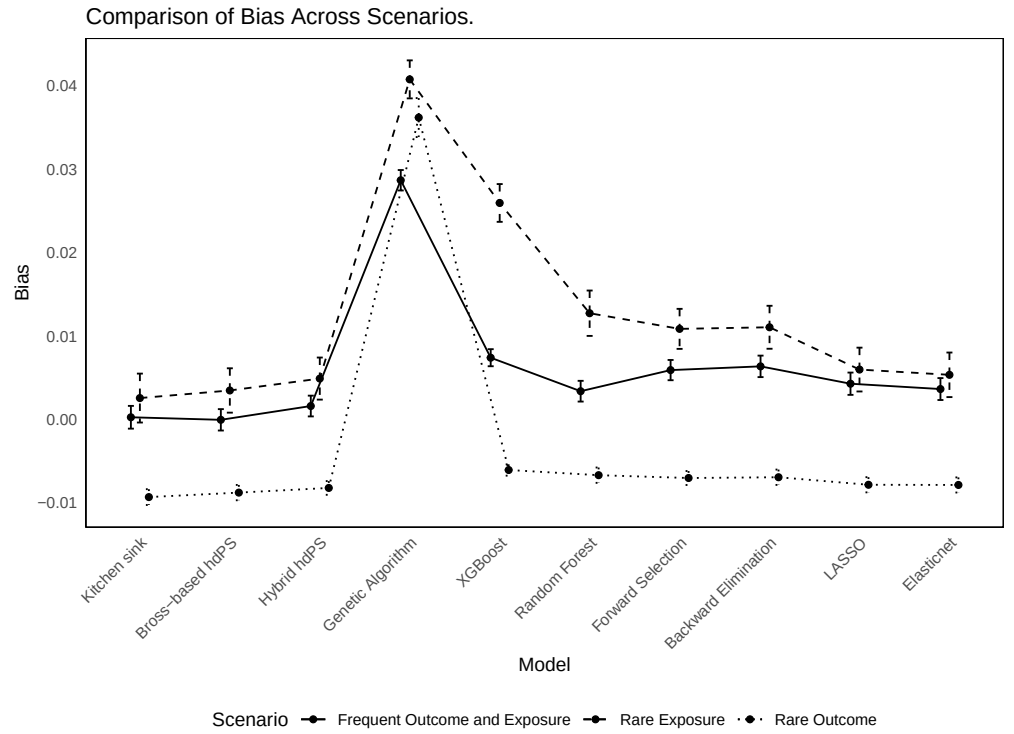


Fig 1. Comparison of Bias Across Different Methods in hdPS Analysis

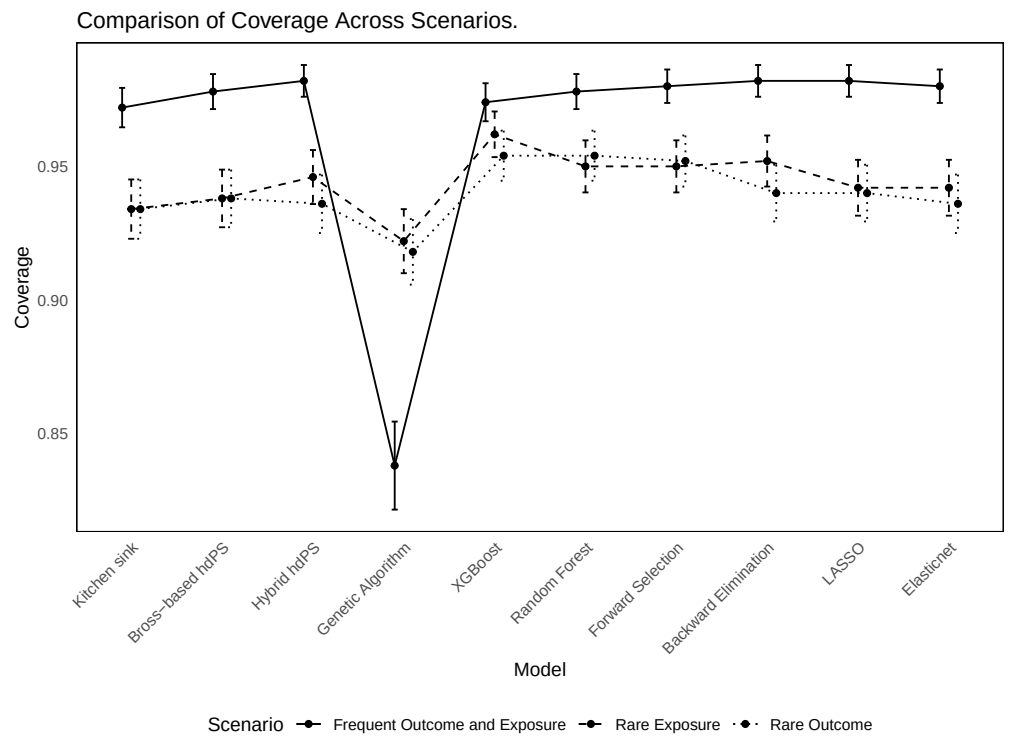


Fig 2. Comparison of Coverage Probability Across Different Methods in hdPS Analysis

variability. LASSO (0.0299) and Elastic Net (0.0294) had slightly higher variability. The Model-based SE followed a similar pattern, with XGBoost (0.0268) showing the lowest variability and the kitchen sink model (0.0333) the highest, indicating less precision in its estimates. When comparing relative error in SE estimation, XGBoost performed the worst. See Appendix §C for further details.

(ii) Rare Exposure and Frequent Outcome:

1. *Bias*: The kitchen sink model showed a relatively low bias (0.0025), while Genetic algorithm continued to exhibit the highest bias (0.0408), indicating a significant deviation from the true effect. XGBoost had a bias of 0.0259, which, while still higher than other methods, but was lower than Genetic algorithm. Other methods, such as Bross-based hdPS (0.0035), Hybrid hdPS (0.0049), and Elastic Net (0.0053), demonstrated moderate bias. Random Forest (0.0127) and Forward Selection (0.0108) had slightly higher bias but remained within an acceptable range.
2. *Coverage*: Coverage levels remained high for most methods, with XGBoost achieving the highest coverage at 96.2%, indicating well-calibrated confidence intervals despite its higher bias. The Genetic algorithm method had lower coverage (92.2%), suggesting that its confidence intervals might be narrower, potentially missing the true effect. Other methods such as RF, Forward Selection, Backward Elimination, and Hybrid hdPS maintained coverage values around 94-95%, suggesting adequate interval calibration. Bias-eliminated coverage for Genetic algorithm improved to 94.2%, but it was still slightly lower than other methods (e.g., forward selection).
3. *MSE*: Forward selection demonstrated the lowest MSE (0.0030), and then Hybrid hdPS and XGBoost. The Genetic algorithm method had a higher MSE (0.0043), reflecting its substantial bias and variability. The kitchen sink model also had an MSE of 0.0043, similar to Genetic algorithm, while other methods such as Bross-based hdPS (0.0035), RF (0.0039), and Elastic Net (0.0036) exhibited moderate MSE values, indicating reasonable accuracy.
4. *SE*: The lowest Empirical SE was observed with XGBoost (0.0507) and Genetic algorithm (0.0510), reflecting high precision despite their higher bias. The kitchen sink model had the highest Empirical SE (0.0656), indicating greater variability. Hybrid hdPS (0.0564), Bross-based hdPS (0.0595), and RF (0.0609) showed moderate variability. Forward Selection (0.0537) and Backward Elimination (0.0576) had lower variability compared to the kitchen sink model. In terms of Model-based SE, XGBoost (0.0531) and Genetic algorithm (0.0533) continued to show low variability, while the kitchen sink model had the highest Model-based SE (0.0623), indicating less precision in its estimates. When comparing relative error in SE estimation, XGBoost and kitchen sink model performed the worst (in other direction).

(iii) Frequent Exposure and Rare Outcome:

1. *Bias*: In this scenario, the kitchen sink model exhibited a moderate negative bias (-0.0093), similar to the Bross-based hdPS method (-0.0088). Genetic algorithm showed a significantly higher bias (0.0362), indicating a substantial deviation from the true effect. Among other methods, XGBoost demonstrated the lowest bias (-0.0061), while methods such as Hybrid hdPS (-0.0082), Forward Selection (-0.0070), and Backward Elimination (-0.0070) had slightly higher but still moderate biases. Elastic Net and LASSO both had biases of -0.0079, reflecting slightly larger deviations compared to XGBoost but still within acceptable limits.

2. *Coverage*: Most methods achieved good coverage, with XGBoost, RF, and Forward Selection each achieving a coverage rate of 95.4%, indicating well-calibrated confidence intervals. The Genetic algorithm method, however, had slightly lower coverage (91.8%), indicating that its confidence intervals might be narrower, potentially excluding the true effect. Bross-based hdPS and the kitchen sink model had slightly lower coverage values of 93.8% and 93.4%, respectively. After accounting for bias, the bias-eliminated coverage for most methods, except Genetic algorithm, remained high, with values ranging from 98.4% to 99.0%, indicating that most methods effectively adjusted for bias in their coverage estimates. Genetic algorithm's bias-eliminated coverage was lower at 93.4%, reflecting its higher inherent bias.
3. *MSE*: XGBoost exhibited the lowest MSE (0.0003). Genetic algorithm had the highest MSE (0.0040), reflecting its substantial bias and variability. The kitchen sink model (0.0005), Bross-based hdPS (0.0005), and other methods such as Hybrid hdPS (0.0004) and Elastic Net (0.0005) all had relatively similar MSE values, indicating moderate accuracy.
4. *SE*: The lowest Empirical SE was observed with XGBoost (0.0152), reflecting high precision in its estimates. The Genetic algorithm method exhibited the highest Empirical SE (0.0523), indicating greater variability and less precision. Methods such as Hybrid hdPS (0.0184), Forward Selection (0.0187), and Elastic Net (0.0203) showed moderate variability, while Bross-based hdPS (0.0206) and the kitchen sink model (0.0212) had slightly higher variability. In terms of Model-based SE, XGBoost (0.0179) again showed the lowest variability, consistent with its low Empirical SE, indicating that it provided the most stable estimates. The kitchen sink model had a slightly higher Model-based SE (0.0219), indicating less precision in its estimates. When comparing relative error in SE estimation, XGBoost performed the worst.

Real-world analysis

Summary results: The dataset comprises 7,585 individuals. Among these, the prevalence of the exposure is 48.8%, while the prevalence of the outcome is 23.7%.

See Fig 3 for the results from analyzing the NHANES (2013-2018) dataset. The methods are arranged according to the number of selected proxy variables. Among all variable selection algorithms, Forward Selection and LASSO yield the highest ORs, with values of 1.56 and 1.55, respectively. The ORs for the remaining methods cluster around 1.50 to 1.54. Additionally, with the exception of a few methods such as Kitchen Sink and Genetic Algorithm, a general pattern emerges in which methods selecting a smaller number of proxy variables tend to produce higher ORs. For RD, Forward Selection again shows the highest value at 0.082, followed closely by LASSO (0.080). Most other methods yield RD values between 0.075 and 0.079, reflecting a similar pattern to that observed for the ORs. This consistency suggests that the relationship between variable selection method and effect size persists across different estimands.

Table 2 presents a pairwise comparison of the number of proxy features shared between different variable selection methods used in the analysis. Each cell in the table indicates the count of common proxy variables selected by the method in the corresponding row and column. The diagonal cells, where the row and column methods are the same, represent the total number of proxy variables selected exclusively by each method. The first column displays the number of proxy variables shared between each method and the kitchen sink (KS) model. Given that the KS model includes all proxy variables, the values in the first column are identical to those in the diagonal cells for

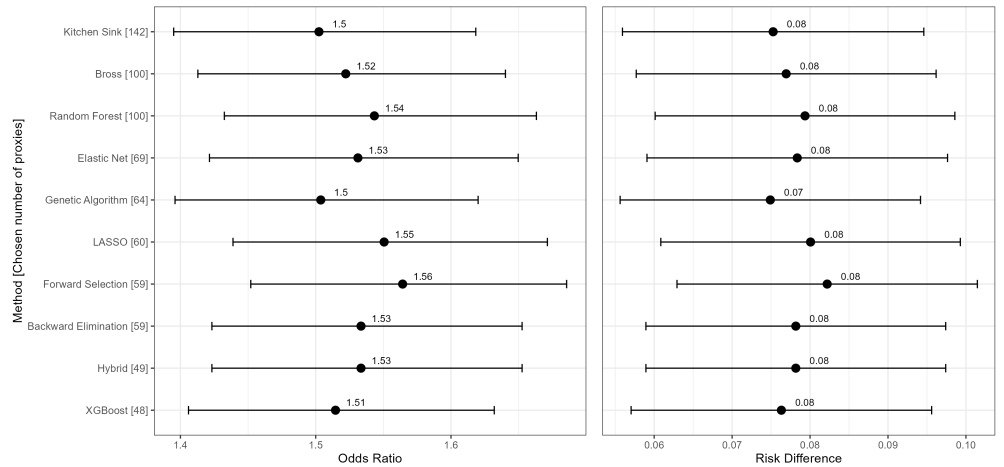


Fig 3. Figure presenting a comparison of Risk Differences (RD) and Odds Ratios (OR) with 95% confidence intervals for different methods used to evaluate the association between obesity and diabetes risk. The analysis is based on data from the National Health and Nutrition Examination Survey (NHANES) for the years 2013-2018. Methods are arranged by the number of variables used in the models.

each row, which also presents the total number of proxy variables selected by the method in the corresponding row.

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Table 2. Comparison of variable overlap of selected proxies across different methods used to evaluate the association between obesity and diabetes from the National Health and Nutrition Examination Survey (NHANES) for the years 2013-2018. **Diagonal entries show the total number of proxies selected by each method, and off-diagonal entries represent the count of shared variables between method pairs.** Most methods share a moderate number of proxies (typically 50–60 percent of the smaller set), indicating partial agreement in variable selection. Higher overlap is observed between closely related methods (e.g., LASSO and Elastic Net, or Hybrid with Bross/LASSO), while methods like XGBoost and Genetic Algorithm show lower overlap with others, reflecting divergent selection behavior in high-dimensional settings.

	KS	Bross	Hybrid	LASSO	EN	RF	XGB	FS	BE	GA
Kitchen sink (KS)	142									
Bross formula	100	100								
Hybrid (Bross and LASSO)	49	49	49							
LASSO	60	47	47	60						
Elastic Net (EN)	69	54	48	60	69					
Random Forest (RF)	100	71	38	46	52	100				
XGBoost (XGB)	48	38	24	28	30	37	48			
Forward selection (FS)	59	45	41	51	54	45	25	59		
Backward elimination (BE)	59	45	41	51	54	45	25	59	59	
Genetic algorithm (GA)	64	44	28	36	40	49	25	35	35	64

Table 3 presents a comparison of the count and percentage of common proxy variables selected by different methods in comparison with the Bross formula hdPS. The first column shows the total number of proxy variables selected by each method, while the second column lists the number of common features selected by both the respective method and the Bross formula. The third column reports the percentage of common features out of the total number of features selected by each method.

We observe that, aside from the Bross formula hdPS and Random Forest models, where the number of proxies was manually set to 100, and the kitchen sink (KS) model, which includes all proxies by design, the number of proxy variables selected by other models ranges between 49 and 69. As expected, the hybrid method combining Bross and LASSO hdPS selects exactly 49 proxy variables, all of which are selected by both methods, resulting in a common feature rate of 1.00. For other models, the common feature percentage is generally clustered around 74%, with XGBoost showing the highest common percentage at 79%, while GA displays the lowest common percentage at 69%.

Table 3. Comparison of the count and percentage of proxy variables selected by each methods in common with that by the Bross formula-based high-dimensional propensity score to evaluate the association between obesity and diabetes from the National Health and Nutrition Examination Survey (NHANES) for the years 2013-2018. **The Hybrid method (Bross + LASSO) shows perfect agreement with the Bross-based hdPS by design. Other methods demonstrate overlap rates ranging from 0.69 to 0.79, indicating moderate consistency in variable selection. XGBoost shows the highest overlap (79 percent) among the non-hybrid methods, while the Genetic Algorithm shows the lowest (69 percent), reflecting greater divergence in selected proxies. These overlap patterns highlight methodological differences in how each approach prioritizes covariates in high-dimensional settings.**

Method	Total Count	Common Count	Rate in Common
Kitchen sink	142	-	-
Bross formula	100	-	-
Hybrid (Bross and LASSO)	49	49	1.00
LASSO	60	47	0.78
Elastic Net	69	54	0.78
Random Forest	100	71	0.71
XGBoost	48	38	0.79
Forward selection	59	45	0.76
Backward elimination	59	45	0.76
Genetic algorithm	64	44	0.69

Computing time:
 Fig 4 presents the computing time for each method. All methods, aside from RF and GA, exhibit relatively fast computing times. RF and GA are generally much slower.

Discussion

Summary of the simulation findings

Comparison of methods:

Across the three scenarios, XGBoost consistently achieved the lowest MSE and high coverage, making it one of the most reliable methods in terms of precision. However, it consistently exhibited some degree of bias, particularly when compared to methods such

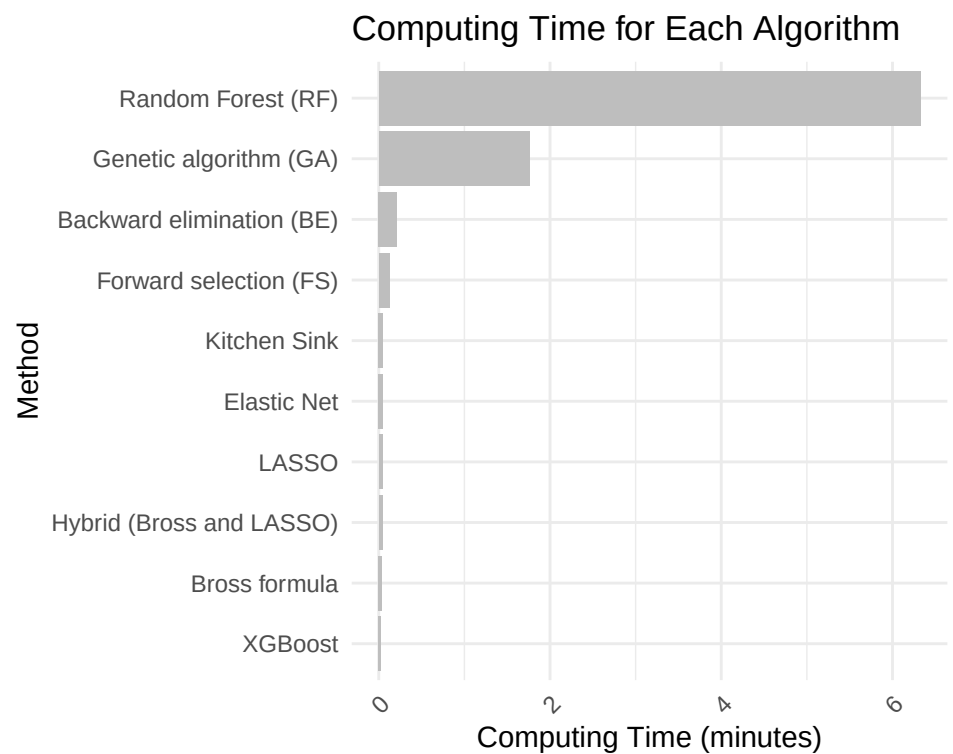


Fig 4. Computing time for the real-world analysis for each algorithm under consideration. The analysis is based on data from the National Health and Nutrition Examination Survey (NHANES) for the years 2013-2018.

as the kitchen sink model, Bross-based hdPS, and Hybrid hdPS, which often showed lower bias in scenarios with frequent outcomes. In contrast, GA displayed the highest bias and MSE, along with lower coverage and greater variability, making it the least reliable method for accurate effect estimation. Methods such as Bross-based hdPS, Hybrid hdPS, and Elastic Net performed moderately well across all scenarios, balancing bias, coverage, and MSE. However, these methods did not outperform XGBoost in terms of overall precision, especially with respect to MSE, though they often resulted in lower bias. The kitchen sink model performed comparably to Bross-based hdPS in terms of bias and coverage but lagged in SE estimation and MSE.

While XGBoost consistently yielded the lowest MSE across scenarios—indicating high overall accuracy—this gain came at the cost of increased bias, particularly in rare exposure settings. This pattern suggests that while XGBoost is highly efficient in minimizing prediction error, it may over-prioritize variance reduction at the expense of bias, potentially due to aggressive regularization or suboptimal feature weighting in the context of non-linear, sparse exposure structures. In causal inference applications, such trade-offs warrant caution, as models with high precision but biased estimates can result in systematically misleading conclusions regarding treatment effects.

GA exhibited consistently high bias and MSE, indicating substantial limitations in handling the dimensionality and noise characteristics inherent in the hdPS framework. One plausible explanation is that GA’s stochastic search process lacks the structure to efficiently navigate large covariate spaces with many weakly informative or correlated variables. Unlike regularized regression or tree-based methods, GA does not incorporate intrinsic mechanisms for managing multicollinearity or prioritizing marginally predictive features, which may lead to unstable or suboptimal variable subsets and subsequently poor confounding adjustment. Its computational intensity further compounds these limitations, making it less practical for high-dimensional epidemiologic analyses.

Comparison of scenarios:

In scenarios with rare exposure, higher bias was observed, particularly for methods such as GA and XGBoost, whereas frequent outcomes generally led to lower bias across most methods. Overcoverage was more common in the scenario with frequent exposure and outcome, with several methods producing confidence intervals that were too wide, suggesting an overestimation of uncertainty. The other scenarios exhibited more balanced or slightly under-coverage. Scenarios with frequent exposure also displayed higher relative error in SE estimation, making it more difficult to precisely estimate effects. In contrast, rare exposure scenarios were associated with higher MSE, reflecting the challenge of estimating effects when the exposure is uncommon. Rare outcome scenarios exhibited the lowest MSE across most methods (except GA), indicating that these scenarios provided a better balance between bias and precision for most methods.

These findings have practical implications for selecting variable selection strategies in epidemiological research. For studies focused on rare exposures (e.g., occupational or genetic risk factors), methods that minimize bias—such as Bross-based hdPS, Hybrid hdPS, or forward/backward stepwise selection—may be more suitable, as they provide more stable effect estimates under sparse data conditions. In contrast, studies targeting common conditions or aiming to improve precision, such as evaluations of widely used treatments or lifestyle exposures, may benefit from machine learning method such as XGBoost, which consistently demonstrate lower MSE. However, trade-offs between bias and variance should be carefully considered in relation to the study’s primary inference goal.

Contextualizing the literature

Previous studies have shown that LASSO performs well within the hdPS framework, particularly in terms of bias reduction and MSE [7,9], with similar results for Elastic

Net [7]. Our findings largely support these conclusions, as both methods demonstrated moderate bias and MSE across the scenarios. However, their performance was inconsistent in some cases, with higher bias observed in rare exposure scenarios, as noted in previous studies [7,9]. MSE also increased in rare exposure scenarios. The Hybrid hdPS method, combining Bross-based hdPS and LASSO, showed promising results, especially in terms of MSE, suggesting it may serve as a suitable alternative to traditional hdPS methods. Random Forest also performed similarly in our study, yielding relatively low bias, which is consistent with previous findings [7]. However, none of the earlier works emphasized coverage or examined methods such as GA, XGBoost, forward selection, or backward elimination, which were evaluated here.

Data analysis findings

The real-world dataset, with frequent exposure (48.8%) and a moderate outcome rate (23.7%), produced ORs of 1.56 and 1.55 for Forward Selection and LASSO, respectively, while other methods clustered around OR values between 1.50 and 1.54. A general trend emerged, showing that methods selecting fewer proxy variables tended to yield higher ORs, with exceptions such as the Kitchen Sink and Genetic Algorithm, which selected more proxies but produced lower ORs. In terms of risk difference (RD), estimates were relatively stable across methods, with Forward Selection again yielding the highest RD (0.082), followed by LASSO (0.080), while others ranged from 0.075 to 0.079. Regarding variable overlap, most methods shared around 74% of their proxy variables with the Bross formula hdPS, with XGBoost showing the highest common rate (79%) and GA the lowest (69%). Computing time analysis showed that, aside from RF and GA, most methods had relatively fast computing times, with RF being significantly slower.

In applied research, particularly with large-scale electronic health records or claims data, computational efficiency is a critical consideration. While tree-based or stochastic optimization methods may offer performance gains, their longer runtime may be impractical in settings requiring rapid analyses or model updates. Simpler approaches such as stepwise selection or penalized regression offer viable alternatives when speed, transparency, or reproducibility are prioritized.

Strengths

Previous studies on singly robust methods have mainly focused on hdPS performance using MSE as the primary evaluation metric [7,9–12]. Our study extends this body of work by incorporating a broader range of performance metrics, allowing researchers to compare results in terms of both bias and variance estimation. This comprehensive comparison of statistical and machine learning methods across various scenarios has not been conducted before. For instance, we found that while XGBoost consistently demonstrated strong performance in terms of MSE and coverage, it did not always have the lowest bias. Additionally, we observed that traditional variable selection methods such as forward selection and backward elimination performed similarly to more sophisticated methods such as LASSO and Random Forest across scenarios, both in terms of bias and coverage.

Our study also employed a complex plasmode simulation framework, closely replicating real-world data conditions [18]. This framework not only accounted for model misspecification, where the true relationships between covariates and outcomes were unknown, but also introduced noise variables, addressing the challenge of dealing with irrelevant covariates in high-dimensional settings. By applying transformations to continuous variables and adding proxy variables that contributed minimally to the outcome, we rigorously evaluated how well each method handled both model

uncertainty and non-informative variables, further strengthening the real-world
relevance of our findings.

Future Direction

Double robust methods have demonstrated strong potential for achieving optimal
statistical performance in hdPS analyses [10,27,28]. In addition, single robust methods,
such as ensemble learners such as super learners, have shown promise in improving bias,
MSE, area under the curve (AUC), and covariate balance in other contexts [11,29,30].
Deep learning methods, particularly supervised architectures, offer significant potential
for improving propensity score estimation in high-dimensional settings by capturing
complex, non-linear relationships and performing well in scenarios with rare exposures,
while maintaining comparable bias and superior variance estimation compared to
traditional methods [31]. Despite their theoretical advantages, the complexity and
computational demands of these methods, especially in high-dimensional settings, have
limited their adoption by practitioners. On the other hand, the simpler singly robust
machine learning methods evaluated here have been applied in clinical research [32,33].

Future research should explore the application of double robust methods and super
learners in hdPS analyses, particularly for handling rare outcomes and exposures, where
the performance of traditional methods may be suboptimal. Additionally, investigating
the impact of different hyperparameters on machine learning methods such as XGBoost
could optimize their performance in hdPS analysis. Finally, future simulation studies
should focus on evaluating coverage and other metrics in epidemiological scenarios such
as time-varying exposures, multiple treatment settings, and data with substantial
measurement error, which would provide valuable insights into the generalizability of
our findings [34].

One avenue for improving bias reduction in machine learning models within hdPS
analysis is the incorporation of causal structure or prior knowledge into variable
selection. Methods such as targeted maximum likelihood estimation, double machine
learning, or feature engineering based on directed acyclic graphs can help align machine
learning-based selection with the underlying causal model. Additionally, tuning
hyperparameters to favor less aggressive regularization or ensemble weighting may
reduce over-smoothing and improve bias performance, particularly in low-prevalence
strata.

Hybrid machine learning models—such as combinations of domain-informed methods
(e.g., Bross or investigator-specified variables) with data-driven algorithms like LASSO
or Random Forest—offer a promising balance between interpretability and predictive
accuracy. Their application in health data science could enable more robust causal
inference while accommodating real-world data challenges such as missingness,
nonlinearity, and high dimensionality. Such models may help mitigate confounding bias
without discarding variables of known clinical relevance, improving both validity and
stakeholder trust in observational research findings.

Conclusion

In conclusion, this analysis highlights the importance of carefully selecting appropriate
methods for hdPS analysis based on the specific characteristics of the data, particularly
the prevalence of exposure and outcome. These findings also emphasize the need to
tailor method selection to the specific epidemiological scenario, ensuring that the chosen
method aligns with the study’s goals, whether minimizing bias or maximizing precision.

XGBoost consistently demonstrated strong performance in terms of MSE and
coverage, making it an effective choice when precision is prioritized. However, it did not
achieve the lowest bias, particularly in rare exposure scenarios, indicating that it may

be less suited when minimizing bias is the primary objective. In contrast, the GA exhibited significant limitations, with consistently high bias and MSE, making it less reliable for effect estimation.

The kitchen sink, Bross-based hdPS, and Hybrid hdPS methods provided a more balanced approach, offering low bias and moderate MSE, though coverage varied depending on the scenario. The analysis also revealed that rare outcomes were associated with lower MSE and better precision, while rare exposures presented challenges, yielding higher bias and MSE. Interestingly, traditional statistical approaches such as forward selection and backward elimination performed comparably to more sophisticated machine learning methods across many scenarios. This suggests that simpler approaches can still be viable, particularly in terms of bias and coverage, and might be preferred due to their computational efficiency.

In real-world healthcare settings—such as electronic health record-based cohort studies, claims data analysis, or comparative effectiveness research—these results offer concrete guidance on choosing hdPS-compatible variable selection strategies. For instance, institutions conducting routine surveillance or drug safety studies may benefit from hybrid or regularized regression approaches that balance scalability with confounding control. Meanwhile, traditional stepwise methods remain attractive for smaller studies or settings requiring transparent modeling. Overall, aligning method choice with data availability, computational resources, and inferential goals can improve the practical utility of hdPS analyses in healthcare.

List of abbreviations

- hdPS: High-dimensional Propensity Score
- NHANES: National Health and Nutrition Examination Survey
- OR: Odds Ratio
- RD: Risk Difference
- SE: Standard Error
- MSE: Mean Squared Error
- KS: Kitchen Sink
- LASSO: Least Absolute Shrinkage and Selection Operator
- EN: Elastic Net; a regularized regression method that combines LASSO and Ridge regression
- RF: Random Forest
- XGBoost: Extreme Gradient Boosting
- FS: Forward Selection
- BE: Backward Elimination
- GA: Genetic Algorithm
- CV: Cross-Validation
- RR: Relative Risk

Declarations

Acknowledgment

This research was supported in part through computational resources and services provided by Advanced Research Computing at the University of British Columbia.

Consent for publication

The National Health and Nutrition Examination Survey (NHANES), conducted by the U.S. Centers for Disease Control and Prevention (CDC), involves collecting data through direct physical examinations, laboratory testing, and interviews. The CDC already obtains consent from participants when collecting this data. When researchers use NHANES data for their studies, they are typically using de-identified, publicly available data. This means that the information cannot be linked back to individual participants, and therefore, additional consent from participants is not required for researchers to use this data.

Availability of data and materials

NHANES data is publicly accessible and can be retrieved from the NHANES website. The datasets generated and/or analyzed during the current study are available in the NHANES repository: <https://www.cdc.gov/nchs/nhanes/index.htm>. Analysis data and codes are provided in <https://github.com/ehsanx/hdPS-proxy-select-codes>, with a stable, citable release archived at Zenodo (DOI: 10.5281/zenodo.15201233 and URL: <https://doi.org/10.5281/zenodo.15201233>). Additionally, all simulation results can be interactively reviewed through a Shiny web application at <https://ehsanx.shinyapps.io/hdPS-Alternatives/>.

Clinical trial number

Not applicable.

Competing interests

MEK is currently supported by grants from Canadian Institutes of Health Research and MS Canada. MEK has previously received consulting fees from Biogen Inc. for consulting unrelated to this current work. MEK was also previously supported by the Michael Smith Foundation for Health Research Scholar award. This does not alter our adherence to PLOS ONE policies on sharing data and materials.

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Authors' contributions

MEK: Conceptualization, Writing – Original Draft, Review & Editing YL: Formal Analysis, Review & Editing

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Not applicable.

References

1. VanderWeele TJ. Principles of confounder selection. *European journal of epidemiology*. 2019;34: 211–219.
2. Schneeweiss S, Rassen JA, Glynn RJ, Avorn J, Mogun H, Brookhart MA. High-dimensional propensity score adjustment in studies of treatment effects using health care claims data. *Epidemiology (Cambridge, Mass)*. 2009;20: 512.
3. Wyss R, Fireman B, Rassen JA, Schneeweiss S. Erratum: High-dimensional propensity score adjustment in studies of treatment effects using health care claims data. *Epidemiology*. 2018;29: e63–e64.
4. Bross ID. Spurious effects from an extraneous variable. *Journal of chronic diseases*. 1966;19: 637–647.
5. Karim ME. High-dimensional propensity score and its machine learning extensions in residual confounding control. *The American Statistician*. 2024; 1–38.
6. Wyss R, Yanover C, El-Hay T, Bennett D, Platt RW, Zullo AR, et al. Machine learning for improving high-dimensional proxy confounder adjustment in healthcare database studies: An overview of the current literature. *Pharmacoepidemiology and Drug Safety*. 2022;31: 932–943.
7. Karim ME, Pang M, Platt RW. Can we train machine learning methods to outperform the high-dimensional propensity score algorithm? *Epidemiology*. 2018;29: 191–198.
8. Schneeweiss S, Eddings W, Glynn RJ, Patorno E, Rassen J, Franklin JM. Variable selection for confounding adjustment in high-dimensional covariate spaces when analyzing healthcare databases. *Epidemiology*. 2017;28: 237–248.
9. Franklin JM, Eddings W, Glynn RJ, Schneeweiss S. Regularized regression versus the high-dimensional propensity score for confounding adjustment in secondary database analyses. *American journal of epidemiology*. 2015;182: 651–659.
10. Pang M, Schuster T, Filion KB, Schnitzer ME, Eberg M, Platt RW. Effect estimation in point-exposure studies with binary outcomes and high-dimensional covariate data—a comparison of targeted maximum likelihood estimation and inverse probability of treatment weighting. *The international journal of biostatistics*. 2016;12.
11. Wyss R, Schneeweiss S, Van Der Laan M, Lendle SD, Ju C, Franklin JM. Using super learner prediction modeling to improve high-dimensional propensity score estimation. *Epidemiology*. 2018;29: 96–106.
12. Simon V, Vadel J. Evaluating the performance of high-dimensional propensity scores compared with standard propensity scores for comparing antihypertensive therapies in the CPRD GOLD database. *Cardiology and Therapy*. 2023;12: 393–408.

13. Saydah S, Bullard KM, Cheng Y, Ali MK, Gregg EW, Geiss L, et al. Trends in cardiovascular disease risk factors by obesity level in adults in the united states, NHANES 1999-2010. *Obesity*. 2014;22: 1888–1895. 651
14. Liu J, Hay J, Faught BE, et al. The association of sleep disorder, obesity status, and diabetes mellitus among US adults—the NHANES 2009-2010 survey results. *International journal of endocrinology*. 2013;2013. 652
15. Kabadi SM, Lee BK, Liu L. Joint effects of obesity and vitamin d insufficiency on insulin resistance and type 2 diabetes: Results from the NHANES 2001–2006. *Diabetes care*. 2012;35: 2048–2054. 653
16. Ostchega Y, Hughes JP, Terry A, Fakhouri TH, Miller I. Abdominal obesity, body mass index, and hypertension in US adults: NHANES 2007–2010. *American journal of hypertension*. 2012;25: 1271–1278. 654
17. Greenland S, Pearl J, Robins JM. Causal diagrams for epidemiologic research. *Epidemiology*. 1999; 37–48. 655
18. Franklin JM, Schneeweiss S, Polinski JM, Rassen JA. Plasmode simulation for the evaluation of pharmacoepidemiologic methods in complex healthcare databases. *Computational statistics and data analysis*. 2014;72: 219–226. 656
19. Farley JF, Harley CR, Devine JW. A comparison of comorbidity measurements to predict healthcare expenditures. *American Journal of Managed Care*. 2006;12: 110–118. 657
20. Morris TP, White IR, Crowther MJ. Using simulation studies to evaluate statistical methods. *Statistics in medicine*. 2019;38: 2074–2102. 658
21. White IR, Pham TM, Quartagno M, Morris TP. How to check a simulation study. *International Journal of Epidemiology*. 2023; dyad134. 659
22. Enders D, Ohlmeier C, Garbe E. The potential of high-dimensional propensity scores in health services research: An exemplary study on the quality of care for elective percutaneous coronary interventions. *Health services research*. 2018;53: 197–213. 660
23. Breiman L. Random forests. *Machine learning*. 2001;45: 5–32. 661
24. Chen T, Guestrin C. Xgboost: A scalable tree boosting system. *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining*. 2016. pp. 785–794. 662
25. Xiao Y, Chen Y, Huang R, Jiang F, Zhou J, Yang T. Interpretable machine learning in predicting drug-induced liver injury among tuberculosis patients: Model development and validation study. *BMC Medical Research Methodology*. 2024;24: 92. 663

26. Holland JH. Adaptation in natural and artificial systems. University of Michigan Press; 1975. 677
678
27. Pang M, Schuster T, Filion KB, Eberg M, Platt RW. Targeted maximum likelihood estimation for pharmacoepidemiologic research. *Epidemiology (Cambridge, Mass)*. 2016;27: 570. 679
680
28. Benasseur I, Talbot D, Durand M, Holbrook A, Matteau A, Potter BJ, et al. A comparison of confounder selection and adjustment methods for estimating causal effects using large healthcare databases. *Pharmacoepidemiology and Drug Safety*. 2022;31: 424–433. 681
682
29. Guertin JR, Rahme E, Dormuth CR, LeLorier J. Head to head comparison of the propensity score and the high-dimensional propensity score matching methods. *BMC medical research methodology*. 2016;16: 1–10. 683
684
30. Ju C, Combs M, Lendle SD, Franklin JM, Wyss R, Schneeweiss S, et al. Propensity score prediction for electronic healthcare databases using super learner and high-dimensional propensity score methods. *Journal of Applied Statistics*. 2019;46: 2216–2236. 685
686
31. Karim ME. Can supervised deep learning architecture outperform autoencoders in building propensity score models for matching? *BMC Medical Research Methodology*. 2024;24: 167. 687
688
32. Hossain MB, Johnston JC, Cook VJ, Sadatsafavi M, Wong H, Romanowski K, et al. Role of latent tuberculosis infection on elevated risk of cardiovascular disease: A population-based cohort study of immigrants in british columbia, canada, 1985–2019. *Epidemiology & Infection*. 2023;151: e68. 689
690
33. Basham CA, Karim ME, Cook VJ, Patrick DM, Johnston JC. Post-tuberculosis airway disease: A population-based cohort study of people immigrating to british columbia, canada, 1985–2015. *EClinicalMedicine*. 2021;33. 691
692
34. Neugebauer R, Schmittdiel JA, Zhu Z, Rassen JA, Seeger JD, Schneeweiss S. High-dimensional propensity score algorithm in comparative effectiveness research with time-varying interventions. *Statistics in medicine*. 2015;34: 753–781. 693
694

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Is There a Competitive Advantage to Using Multivariate Statistical or Machine Learning Methods Over the Bross Formula in the hdPS Framework for Bias and Variance Estimation?
PLOS ONE

Authors' response

Please find our point-by-point responses in the boxes.

Reviewer 1

Comment from the Authors: Reviewer 1 kindly provided both in-text comments and a set of rephrased questions in an attached PDF. For clarity and completeness, we have consolidated these comments and listed the corresponding questions below to address them collectively.

The manuscript titled "Is There a Competitive Advantage to Using Multivariate Statistical or Machine Learning Methods Over the Bross Formula in the hdPS Framework for Bias and Variance Estimation?" offers a comprehensive evaluation of various proxy selection methods within the high-dimensional propensity score (hdPS) framework. It compares traditional statistical methods with machine learning approaches, and it effectively assesses their performance across various epidemiological scenarios. The use of a plasmode simulation with NHANES data adds significant value to the study's methodology and conclusions.

Comment from attached PDF:

The paper provides a comprehensive evaluation of various proxy selection methods in the highdimensional propensity score (hdPS) framework. It compares traditional statistical methods with machine learning approaches to assess their performance in terms of bias, mean squared error (MSE), coverage, and standard error (SE) across different epidemiological scenarios. The study also leverages a plasmode simulation using NHANES data to draw conclusions about these methods' efficacy. The findings highlight how machine learning methods like XGBoost outperform traditional methods like the Bross formula in precision, but can incur higher bias in some scenarios. Below are the comments given for each section.

However, I believe that the manuscript requires major revisions to improve its quality and robustness. The following points outline the areas that need attention:

Introduction and Literature Review: The introduction could be enhanced by providing more context on the critical role of hdPS in modern epidemiology. Specifically, expanding on how hdPS addresses the limitations of traditional propensity score methods and directly impacts causal inference in healthcare data would strengthen the manuscript. The literature review could also benefit from a clearer connection to prior research, particularly focusing on how previous studies have applied machine learning to hdPS and how this study brings new insights to the field.

Comment from attached PDF:

- Introduction and literature review
 1. The introduction could benefit from more context on why hdPS is critical in modern epidemiology and how it directly impacts causal inference, especially in large-scale healthcare data analysis.
 2. Consider expanding on the limitations of traditional propensity score methods and how hdPS addresses these challenges.
 3. The literature review section would benefit from a clearer connection to the current study. It could briefly mention how previous work has assessed machine learning in hdPS, setting up the novelty of this study's approach.

Response: In response to the reviewer's suggestion, we have revised the introduction and literature review to better contextualize the role of hdPS in modern epidemiology, clarify its advantages over traditional propensity score methods, and establish how our work builds on existing literature:

"Importantly, hdPS has become a foundational method in modern epidemiologic research that uses healthcare databases, offering a scalable and reproducible approach to confounding control when traditional investigator-specified propensity score models are limited by unmeasured, mismeasured, or omitted variables. Unlike conventional PS methods that rely solely on expert knowledge and a fixed set of covariates, hdPS empirically identifies and prioritizes proxy variables from high-dimensional data, allowing it to adjust for latent confounding structures more effectively. This makes hdPS particularly valuable for improving causal inference in observational healthcare studies, including those evaluating drug safety, treatment effectiveness, and health system interventions [5,6]."

"Several studies have proposed hybrid approaches that combine machine learning and hdPS to enhance variable selection and bias reduction while maintaining interpretability [6,7,9]. ... These efforts reflect a growing interest in refining hdPS with modern statistical learning tools to optimize proxy confounder adjustment in realistic, data-rich settings."

"To extend prior research, our study systematically compares multiple machine learning and hybrid hdPS approaches using plasmode simulations grounded in real-world data structure. Importantly, we expand beyond commonly reported metrics such as bias and MSE by incorporating underutilized diagnostics—namely, empirical coverage and relative error in standard error—to provide a more nuanced understanding of estimator performance."

Methods Section: While the methodology is generally well-detailed, there is a need for further clarification regarding the choice of machine learning models (e.g., Genetic Algorithm, XGBoost) and statistical methods. A better explanation of why these particular models were selected and their relevance to hdPS would make the study's approach more transparent. The "kitchen sink model" is also mentioned briefly but needs further elaboration on its role in the analysis. Furthermore, providing a clearer explanation of the traditional variable selection methods, such as "forward selection" and "backward elimination," would benefit readers who may not be familiar with these techniques.

Comment from attached PDF:

- Methods

4. While the paper provides an extensive methodology section, some aspects of the simulation design could be clearer. For example, explain more clearly the rationale behind selecting the specific machine learning models (e.g., Genetic Algorithm, XGBoost) and statistical methods. What makes these particular models suited for this analysis?
5. The use of the "kitchen sink model" could be elaborated on. It's mentioned as a baseline but lacks a detailed description of its implications in the context of this study.
6. A brief explanation of the "forward selection" and "backward elimination" methods would be beneficial to readers unfamiliar with these techniques, as they are relatively basic compared to machine learning models.

Response: To improve transparency and clarity in our methodological choices, we elaborated on the rationale for selecting specific machine learning models, clarified the role of the kitchen sink model, and expanded explanations of traditional variable selection methods, as follows:

"The selected machine learning models were chosen for their demonstrated ability to handle high-dimensional data and perform feature selection effectively. For example, XGBoost and Random Forest are tree-based ensemble methods known for capturing non-linear relationships and interactions. Genetic Algorithm (GA) offers a stochastic, optimization-based approach to variable selection, allowing exploration of complex covariate spaces. These models align with the goals of hdPS by enabling automated, multivariate proxy variable selection that may capture complex dependencies missed by univariate selection strategies like the Bross formula. Incorporating such models expands the flexibility and potential robustness of hdPS analyses in high-dimensional data settings."

"Kitchen sink model: The kitchen sink model serves as a comprehensive baseline that includes all available covariates—both investigator-specified and proxy variables—without any variable selection. It is used to benchmark the performance of variable selection methods against a model that assumes complete adjustment [7]."

"Stepwise: Stepwise selection is a commonly used variable selection technique that builds or reduces models by adding or removing predictors based on statistical criteria—in our case, adjusted R-squared. We implemented two versions: (a) Forward selection (FS) begins with a minimal model, typically including only investigator-selected covariates. It then adds one proxy variable at a time—the one that most improves model performance at each step. This process continues until no additional variable meaningfully improves the fit. (b) Backward elimination (BE) takes the opposite approach. It starts with a full model containing all investigator-selected and proxy variables and removes the least informative variable at each step, based on its contribution to model fit. This continues until removing more variables no longer improves or begins to harm the model. These stepwise approaches are transparent, making them accessible alternatives for confounding adjustment in high-dimensional settings."

Results and Discussion: The simulation results are well-presented, but additional discussion on the interpretation of bias and MSE would help contextualize the trade-offs involved in using machine learning methods like XGBoost. While XGBoost shows lower MSE, the increase in bias should be explored in more detail to help readers understand the model's limitations. Similarly, the poor performance of the Genetic Algorithm should be explained with greater depth, especially regarding its struggles with high-dimensional data compared to other methods.

Comment from attached PDF:

- Results
- 7. While the simulation results are well-presented, additional discussion on how bias and MSE were interpreted across the different methods would help. For instance, while XGBoost shows lower MSE, the increase in bias could be further explored to contextualize the trade-offs involved.
- 8. The explanation for why Genetic Algorithm performs poorly in terms of bias and MSE should be more detailed. Are there particular reasons why this algorithm struggles with high-dimensional data compared to others?

Response: We expanded the discussion to clarify the interpretation of bias and MSE, particularly to contextualize trade-offs associated with models like XGBoost, and to provide a more detailed explanation of the Genetic Algorithm's limitations:

"While XGBoost consistently yielded the lowest MSE across scenarios—indicating high overall accuracy—this gain came at the cost of increased bias, particularly in rare exposure settings. This pattern suggests that while XGBoost is highly efficient in minimizing prediction error, it may over-prioritize variance reduction at the expense of bias, potentially due to aggressive regularization or suboptimal feature weighting in the context of non-linear, sparse exposure structures. In causal inference applications, such trade-offs warrant caution, as models with high precision but biased estimates can result in systematically misleading conclusions regarding treatment effects.

GA exhibited consistently high bias and MSE, indicating substantial limitations in handling the dimensionality and noise characteristics inherent in the hdPS framework. One plausible explanation is that GA's stochastic search process lacks the structure to efficiently navigate large covariate spaces with many weakly informative or correlated variables. Unlike regularized regression or tree-based methods, GA does not incorporate intrinsic mechanisms for managing multicollinearity or prioritizing marginally predictive features, which may lead to unstable or suboptimal variable subsets and subsequently poor confounding adjustment. Its computational intensity further compounds these limitations, making it less practical for high-dimensional epidemiologic analyses."

Tables and Figures: The readability of Table 2 and Table 3 could be improved by adding brief summaries or interpretations of the key results in their captions. This would help readers quickly grasp the significance of the findings and how they support the broader argument.

Comment from attached PDF:

- Table
- 9. The paper could improve the readability of Table 2 and Table 3 by providing a brief summary or interpretation of key results directly in the table captions. A clearer focus on how these results contribute to the broader argument would enhance the table's value.

Response: To enhance readability and aid interpretation, we revised the captions of Tables 2 and 3 to include key summaries that highlight how the results support the study's broader findings:

“Table 2: ... Diagonal entries show the total number of proxies selected by each method, and off-diagonal entries represent the count of shared variables between method pairs. Most methods share a moderate number of proxies (typically 50–60 percent of the smaller set), indicating partial agreement in variable selection. Higher overlap is observed between closely related methods (e.g., LASSO and Elastic Net, or Hybrid with Bross/LASSO), while methods like XGBoost and Genetic Algorithm show lower overlap with others, reflecting divergent selection behavior in high-dimensional settings.”

“Table 3: ... The Hybrid method (Bross + LASSO) shows perfect agreement with the Bross-based hdPS by design. Other methods demonstrate overlap rates ranging from 0.69 to 0.79, indicating moderate consistency in variable selection. XGBoost shows the highest overlap (79 percent) among the non-hybrid methods, while the Genetic Algorithm shows the lowest (69 percent), reflecting greater divergence in selected proxies. These overlap patterns highlight methodological differences in how each approach prioritizes covariates in high-dimensional settings.”

Conclusions: The conclusions are generally strong, but they could be expanded to provide more practical guidance for future research and real-world applications. Specifically, the authors could discuss which methods might be more appropriate for different types of epidemiological studies, such as those focused on rare diseases versus common conditions.

Comment from attached PDF:

- Conclusions

10. The conclusions are robust but could be expanded to include more practical guidance for future research and applications. For example, in which specific types of epidemiological studies (e.g., rare diseases vs. common conditions) would one method be preferred over another?

Response: In response to the request for more practical guidance, we expanded the conclusion to provide recommendations for method selection across different epidemiological study contexts:

“These findings have practical implications for selecting variable selection strategies in epidemiological research. For studies focused on rare exposures (e.g., occupational or genetic risk factors), methods that minimize bias—such as Bross-based hdPS, Hybrid hdPS, or forward/backward stepwise selection—may be more suitable, as they provide more stable effect estimates under sparse data conditions. In contrast, studies targeting common conditions or aiming to improve precision, such as evaluations of widely used treatments or lifestyle exposures, may benefit from machine learning method such as XGBoost, which consistently demonstrate lower MSE. However, trade-offs between bias and variance should be carefully considered in relation to the study’s primary inference goal.”

“Finally, future simulation studies should focus on evaluating coverage and other metrics in epidemiological scenarios such as time-varying exposures, multiple treatment settings, and data with substantial measurement error, which would provide valuable insights into the generalizability of our findings [33].”

Comment from attached PDF:

- Others questions

11. How can machine learning models be improved for higher bias reduction in hdPS analysis?
12. What is the potential impact of using hybrid machine learning models in health data science?
13. How can the results of this study be applied to real-world healthcare settings?

Response: To address the reviewer's broader questions regarding the implications of our findings, we added the following discussion on improving machine learning models for bias reduction, the potential impact of hybrid models, and real-world applications:

"One avenue for improving bias reduction in machine learning models within hdPS analysis is the incorporation of causal structure or prior knowledge into variable selection. Methods such as targeted maximum likelihood estimation, double machine learning, or feature engineering based on directed acyclic graphs can help align machine learning-based selection with the underlying causal model. Additionally, tuning hyperparameters to favor less aggressive regularization or ensemble weighting may reduce over-smoothing and improve bias performance, particularly in low-prevalence strata.

Hybrid machine learning models—such as combinations of domain-informed methods (e.g., Bross or investigator-specified variables) with data-driven algorithms like LASSO or Random Forest—offer a promising balance between interpretability and predictive accuracy. Their application in health data science could enable more robust causal inference while accommodating real-world data challenges such as missingness, nonlinearity, and high dimensionality. Such models may help mitigate confounding bias without discarding variables of known clinical relevance, improving both validity and stakeholder trust in observational research findings."

"In real-world healthcare settings—such as electronic health record-based cohort studies, claims data analysis, or comparative effectiveness research—these results offer concrete guidance on choosing hdPS-compatible variable selection strategies. For instance, institutions conducting routine surveillance or drug safety studies may benefit from hybrid or regularized regression approaches that balance scalability with confounding control. Meanwhile, traditional stepwise methods remain attractive for smaller studies or settings requiring transparent modeling. Overall, aligning method choice with data availability, computational resources, and inferential goals can improve the practical utility of hdPS analyses in healthcare."

These revisions are essential to enhancing the manuscript's clarity, depth, and impact. A more robust discussion of the trade-offs between methods, along with clearer explanations of model choices, will help solidify the paper's contribution to the field.

Recommendation: Given the importance of the topic, I recommend that the authors address the major revisions outlined above. Once these revisions are made, I believe the manuscript could make a valuable contribution to the literature on hdPS and its applications in health data science.

Thank you for considering my review.

Reviewer 2

The manuscript presents a highly relevant comparison of hdPS variable selection methods, including both traditional statistical and machine learning approaches. The findings are useful for epidemiologists and health data scientists.

Key suggestions for revision:

- Clarify the rationale behind selecting 100 proxy variables in hdPS methods.

Response: To address the reviewer's request, we clarified why 100 proxy variables were selected in the hdPS and related methods:

"This threshold of 100 proxy variables has been used in the literature to balance model complexity and overfitting risk, while ensuring that a sufficient number of proxies are included to capture unmeasured confounding [22]. Our choice reflects a practical trade-off between dimensionality and computational feasibility."

- Expand the discussion on bias correction for coverage probability.

Response: We expanded the discussion to explain how bias correction influenced the interpretation of coverage probability:

"Bias-corrected coverage provides a more accurate reflection of model reliability when bias is present. By adjusting for bias, we isolate the coverage performance attributable to variance estimation alone, allowing clearer comparisons of uncertainty calibration. This correction is crucial for distinguishing methods that produce narrow but biased intervals from those that offer valid inference under realistic misspecification."

- Address concerns regarding MSE interpretation when variance estimation is unstable.

Response: In response to the reviewer's concern about MSE interpretation, we added the following clarification:

"While MSE is a commonly used metric, its interpretation becomes less reliable when variance estimates are unstable, as observed with singly robust machine learning models. In such cases, MSE may mask the trade-off between high variance and low bias (or vice versa), especially when confidence intervals are miscalibrated. Our inclusion of coverage and standard error diagnostics addresses this limitation by providing complementary insights into inference accuracy."

- Discuss the practical implications of computational efficiency for real-world applications.

Response: We have now added a discussion on the practical importance of computational efficiency in applied settings:

"In applied research, particularly with large-scale electronic health records or claims data, computational efficiency is a critical consideration. While tree-based or stochastic optimization

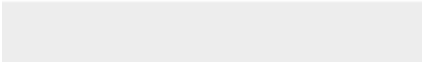
methods may offer performance gains, their longer runtime may be impractical in settings requiring rapid analyses or model updates. Simpler approaches like stepwise selection or penalized regression offer viable alternatives when speed, transparency, or reproducibility are prioritized.”

- Improve clarity and conciseness in the Results and Discussion sections.

Response: We appreciate the suggestion. In line with the other specific comments, we improved clarity by adding interpretive text to the Discussion section (e.g., explaining potential reasons for the direction of results across different methods, model limitations, practical implications, future research directions, and conclusions). These additions aim to enhance readability and help readers more easily interpret the results. All changes are marked in red in the revised manuscript. Given the diversity of simulation scenarios and performance metrics considered, reported results explain the nuance, and comparability of the findings. We hope these revisions adequately address the reviewer’s comment.

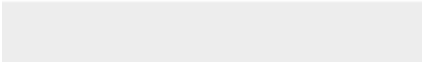


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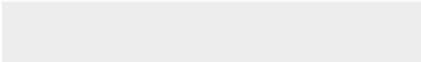


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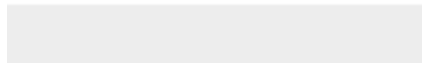
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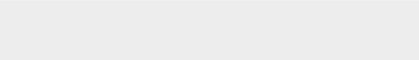
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